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Technology - European Semiconductors | Europe

Time to Be Selective

We remain constructive on semis, underpinned by strong AI demand and a broadening cycle recovery. However, with DRAM approaching a late-cycle inflection and valuation dispersion widening, we turn more selective. We upgrade SNPS to OW, move IFX to EW and retain ASML at OW, but lower our PT to €1,700.

Key Takeaways

- Our coverage has traded up c.70% YTD, but retreated c.60% from its June peaks. We note a slight pull-back in positioning, but interest remains high.
- We highlight 3 key debates per sub-sector:
- Semiconductor Equipment: (i) capacity, (ii) export restrictions, (iii) 2028 WFE growth.
- Analog and Optics: (i) margin recovery, (ii) vertical power delivery, (iii) the multi-rack scale up opportunity.
- Design and IP: (i) LLM threat to EDA, (ii) simulation and physical AI, (iii) the CPU TAM.

More selective within the sector: Synopsys to Overweight, Infineon to Equal-weight, trim PTs for ASML and Besi. On Synopsys, we see an attractive entry point following the recent de-rating, with greater confidence in Ansys synergies and Design IP recovery, alongside an underappreciated opportunity from physical AI; we leave our \$500 PT unchanged. On Infineon, while the structural data center opportunity remains compelling, we see limited near-term upside to expectations, with PSS estimates looking too high and several emerging risks leaving us without a clear catalyst. We leave estimates unchanged but cut our PT to €65 from €81, reflecting a lower multiple for the server rack business. We retain our Overweight rating on ASML, where DRAM and logic demand continue to support the medium-term margin expansion story, but near-term overhangs around China, capacity and margins are likely to constrain the multiple; we leave estimates unchanged and lower our PT to €1,700 from €1,930 on a 30x P/E multiple (at the lower end of the peak cycle multiple range) versus 35x previously. We trim Besi estimates as we view hybrid bonding insertion for HBM as likely to feature later and lower. while also adjusting our multiple to 31x FY27, from 35x previous, resulting in a PT of €220 (from €260).

3 key debates in focus per sub-sector. For our Semiconductor Equipment coverage, we think the talking points are ASML's EUV capacity into 2028, export controls, and whether WFE growth can keep accelerating in 2028. In Analog and Optics, the questions are whether near-term margin recovery can match a bar that keeps moving higher, who wins in Vertical Power Delivery as ADI and Navitas enter a field

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TECHNOLOGY - EUROPEAN SEMICONDUCTORS		
Europe Industry View		In-Line
WHAT'S CHANGED		
ASML Holding NV (ASML.AS)	From	To
Price Target	€1,930.00	€1,700.00
Infineon Technologies AG (IFXGn.DE)	From	To
Rating	Overweight	Equal-weight
Price Target	€81.00	€65.00
Synopsys Inc. (SNPS.O)	From	To
Rating	Equal-weight	Overweight
BE Semiconductor Industries NV (BESI.AS)	From	To
Price Target	€260.00	€220.00

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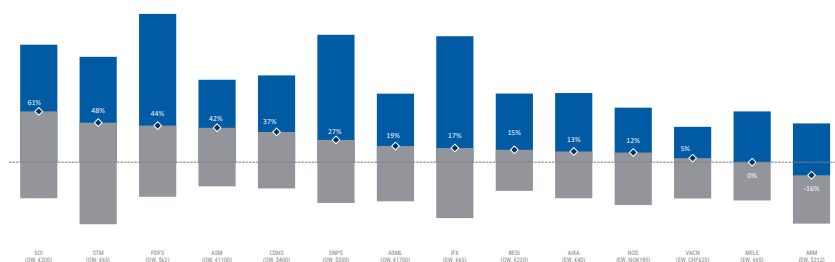
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led by Infineon, Monolithic Power Systems and Renesas, and how multi-rack scale up increases Soitec's revenue opportunity. In EDA/IP, we think focus is on whether the risk of LLMs as a competitive threat is overdone, the emergence of Physical AI, and the CPU TAM.

4 risks into year-end. While we remain constructive on our coverage, we see four key risks that could challenge semis enthusiasm into year-end: (i) **Data center build-out could fall short of elevated expectations**, as grid connection delays and increasing political pushback constrain the pace of incremental GW additions, (ii) **A tougher macro backdrop could pressure sentiment**, with higher long-end yields and oil prices adding to investor scrutiny around NVIDIA's investments across the AI ecosystem, (iii) **The LLM debate could shift from growth to returns**, as investors increasingly question whether falling token prices and improving efficiency are sufficient to support attractive ROICs, and (iv) **Potential delays to 800V-native architecture** could create headwinds to growth, particularly for suppliers whose content uplift depends on the 400V-to-800V transition, including GaN.

Semis in vogue in 2026: a momentary pullback from June peaks, but fundamentals remain constructive. 1H26 saw the semiconductor sector outperform the S&P 500 YTD by over 50% as data center build out and LLM adoption drove sentiment higher for the AI beneficiaries. However, this summer has seen significant volatility, with our stocks de-rating c.60% from June peaks and daily moves in the SOXX having widened to a 4.1% standard deviation from 2.5% over January to May. That said, semis remain among the most-owned sectors, suggesting investor interest remains high, and that the recent sell-off may offer attractive entry points. Fundamentals remain supportive: stronger than expected 1H26 SIA data point to a broadening recovery, firmer pricing and improving demand, reinforcing our expectation for the analog recovery to continue through 2H26. The key wrinkle is memory. DRAM pricing appears close to a second-derivative peak, and our memory analysts expect the cycle to transition to late-cycle by 4Q26. While we think this could weigh on sentiment towards memory-exposed semi-cap equipment names, structural AI tailwinds should provide some cushion. Against this backdrop, valuation dispersions have widened. EDA and several semi-cap names trade below historical multiples, creating opportunities, but we remain selective given emerging cyclical risks.

Exhibit 1: Risk Reward Skew Across Our Coverage



Source: Morgan Stanley Research

Exhibit 2: 3 Key Debates into Year End by Sub-Sector

Semiconductor Equipment	Analog and Optics	EDA and IP
1. Does ASML have the capacity to support WFE demand into 2028?	1. Can margin recovery keep pace with market expectations?	1. Are LLMs a credible threat to EDA?
2. Are export restrictions a risk for Semi-Cap?	2. Who are the beneficiaries of the transtion to Vertical Power Delivery (VPD)?	2. Is 2027 where we will begin to see earnings upside from simulation?
3. Can WFE growth continue to accelerate in 2028?	3. How does multi-rack scale-up increase Soitec's revenue opportunity per GPU?	3. How large is the CPU TAM and who benefits?

Source: Morgan Stanley Research

AlphaSignals Catalyst Event Preview

Company / Catalyst	Date	Importance	Surprise
Synopsys Inc. SNPS.O			
Investor Day	30 Sep 2026	High	↑ Modest upside surprise

Source: Company data, Morgan Stanley Research

Order of Preference

Within Semi-Cap Equipment we prefer our Top Pick ASML (OW, PT €1700) as well as ASM (OW, PT €1100) and Besi (OW, PT €220). Within Analog and Optics we prefer our Top Pick Soitec (OW, PT €200) into 2027 and STMicroelectronics (OW, PT €65). Within EDA and IP we prefer our EDA coverage Cadence (OW, PT \$400) and Synopsys (OW, PT \$500). We initiate on PDF Solutions at an OW rating, PT \$62.

Exhibit 3: Our Semi-Cap Equipment Relative Preferences

Semi-Cap Equipment



OW, PT €1700

Top Pick

ASML is a dominant supplier of critical toolsets for the semi manufacturing process. We expect WFE growth, although we see lithography intensity (share of WFE) flattening over 2024-26, which may cap earnings in the meantime. We anticipate consensus earnings revisions as debate shifts to 2026-27 on improving DRAM and logic demand for EUV. We anticipate better than expected customer demand for holistic lithography solutions. Metrology margins look robust as use of eBeam drives growth and near universal adoption. We expect supply chain management to deliver growing GM% across the mid to longer term.



OW, PT €1100

Leading edge foundry capex is inflecting sharply higher, which should increase supply in the face of heavy AI demand. ASMI has highest exposure to this and should see continued share of wallet gains off every capex dollar spent. We see strong growth for the years ahead.



OW, PT €220

The set-up for Besi's core business continues to improve, evidenced by a need for chiplet architectures (2.5D) for AI accelerators. Accelerating hybrid bonding adoption in AI (both compute and networking) has started to result in meaningful orders. Valuation in the high 20s FY27e P/E is not reflecting optionality of broadening hybrid bonding adoption outside of HBM.



PASSION. PRECISION. PURITY.

EW, PT 625 CHF

VAT is positioned to benefit from strong WFE growth, but we see better value elsewhere and stay EW.



EW, PT €40

Although we are (increasingly) optimistic about the outlook for Opto we see some risk of duration beyond 2027. GaN offers a credible medium-term leg of growth, though near-term pacing remains uneven as customers calibrate investment. The stock is fully valued on c.37x FY27e EPS.

Source: Morgan Stanley Research

Exhibit 4: Our Analog and Optics Relative Preferences

Analog and Optics



OW, PT €200

Top Pick

Soitec is leveraged to silicon photonics (SiPho) through its leadership in photonics SOI substrates, a critical input for optical networking modules. As SiPho adoption scales in AI datacenters, Soitec's opportunity increases. With near term acceleration confirmed on both better demand visibility and execution we see Soitec as a core holding for investors that want exposure to optical networking.



life.augmented

OW, PT €65

STM has a portfolio positioned for growth areas (BiCMOS imaging sensors, GP MCUs). We see opportunity for recovery into FY26/27.

We expect drivers of FY27 growth to come from data centres, with a material ramp in optical revenues driven by pluggables and NPOs (2027+), as well as LEOs in the CECP end market.

We allow for a modest industrial recovery cycle starting 2H26.

We see some signs of automotive recovery as ASIL-grade microcontroller (MCUs) design wins start to convert to sales (2027+).



EW, PT €65

We view Infineon as best placed amongst European analog names for a cyclical recovery in automotive/industrial, demonstrating a superior ability to weather the headwinds facing these markets in the meantime.

We expect new wins in autos to validate IFX's strategy (design focus, capacity build, proprietary IP). A strong IP library augurs well for GaN longer term too including use in EVs, data centres and energy storage.

We think Infineon will be the clear data center beneficiary with the full suite of power semi solutions for each stage of the rack.

However near-term share price movements will depend on clarification of the 800v deployment timeline.



EW, PT €65

Melexis enters 2H26 with higher revenue expectations, but growth still trails peers despite a weaker first-half base. Margin conversion remains muted as management prioritises volume over pricing and wage inflation absorbs gross margin gains, while robotics remains too small to affect the near-term earnings path. We remain EW pending clearer evidence of sustained catch-up growth and stronger operating leverage.

NORDIC[®]
SEMICONDUCTOR

EW, PT 185 NOK

Nordic is well-positioned as a market leader in one of the fastest-growing segments of the IoT market, with recent strong earnings momentum driving a premium valuation. We believe the stock is currently pricing in over 20% growth and EBITDA margins of 25% (= company guidance), achievable, but dependent on: 1) successful launch of the nRF54 series platform, 2) further expansion through deeper penetration of industrial and long-range end markets, 3) disciplined opex control, and 4) effective monetization of recent software acquisitions. Although momentum in all 4 is currently evident we see this embedded in the current valuation.

Source: Morgan Stanley Research

Exhibit 5: Our Design and IP Relative Preferences**Design and IP**


OW, PT \$400

We think the range of growth drivers and the operational efficiency inherent in Cadence's business model has yet to be fully reflected in the share price. The company has a solid base in systems analysis and simulation work in the chip design flow, deep customer relationships, a leading position in analog and mixed-signal design, a solid balance sheet and should enjoy strong structural growth with its new AI-driven EDA tools. With our FY27 EPS of \$9.71 and our view that the business should trade on c.40x earnings, we are Overweight with a \$400 price target.



OW, PT \$500

We think opportunities in systems simulation present a multi-year growth driver. We assume a 1-year forward P/E multiple range of 30-35x in setting our price target of \$500. We are Overweight as we think (i) Design IP recovery is becoming more evident, (ii) Ansys synergies are materialising and (iii) the simulation portfolio is positioned to benefit from the emergence of physical AI.



OW, PT \$62

PDFS offers differentiated exposure to semis manufacturing, with structural complexity, a predominantly recurring revenue base and emerging AI monetisation supporting durable growth and margin expansion. We see backlog conversion, gross-margin recovery and tangible AI adoption are the key proof points for further upside.



EW, PT \$212

Arm is the silicon IP leader with 60%+ share of global semiconductor IP sales, including a ubiquitous position in smartphones. Upside revolves around structural growth opportunities in edge/cloud AI, PCs & autos including share gain, royalty rate inflections as customer designs shift to Arm's v9/CSS architecture and upside from the chip transition. We model a c.31% EPS CAGR over FY24-28 amidst the shift to chipmaking. Downside risks include outstanding litigation, geopolitical uncertainty, competition and slowing future China sales.

Source: Morgan Stanley Research

Positioning and Sector Outlook

Within our coverage, European stocks are up +88% YTD, while US-listed EDA and IP companies are relatively flat. Although our sector remains relatively well owned, multiples look attractive given recent de-grossing, which keeps us constructive on the sector. We continue to see AI demand driving growth for our stocks but note key risks emerging vs the start of the year (see our [2026 outlook](#)) as the debate shifts to 2028. We maintain our preference for ASML, but now also reiterate our Top Pick Soitec.

A momentary pull back from June peaks?

Our coverage has traded up c.70% YTD, but retreated c.60% from its June peaks.

While all names across our European semiconductor coverage have traded up meaningfully this year, the timing of the price action has differed by sub-sector. Q1 activity remained concentrated in semi-cap equipment, outperforming analog and optics by 14%. However, activity shifted in Q2 to analog and optics stocks, which traded up 97% in the quarter, outperforming semi-cap by 47%. Overall, 1H26 market sentiment remained relatively bullish, with European semiconductors broadly rewarded as AI enablers. However, this summer has seen significant volatility, with our stocks de-rating c.40% from June peaks and daily moves in the SOXX having widened to a 4.1% standard deviation from 2.5% over January to May. Performance across our US-listed coverage was more mixed this year, with our EDA coverage underperforming the SOX (CDNS -3% YTD and SNPS -11% YTD) as LLM risk to EDA overshadowed strong financials. Arm rallied 122% YTD on the launch of the Arm AGI CPU.

Exhibit 6: Price performance of our coverage YTD (%)



Exhibit 7: The SOX has outperformed the S&P 500 by more than 50% YTD.

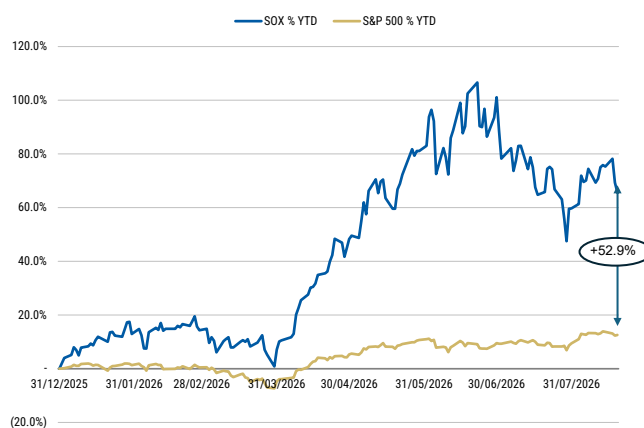
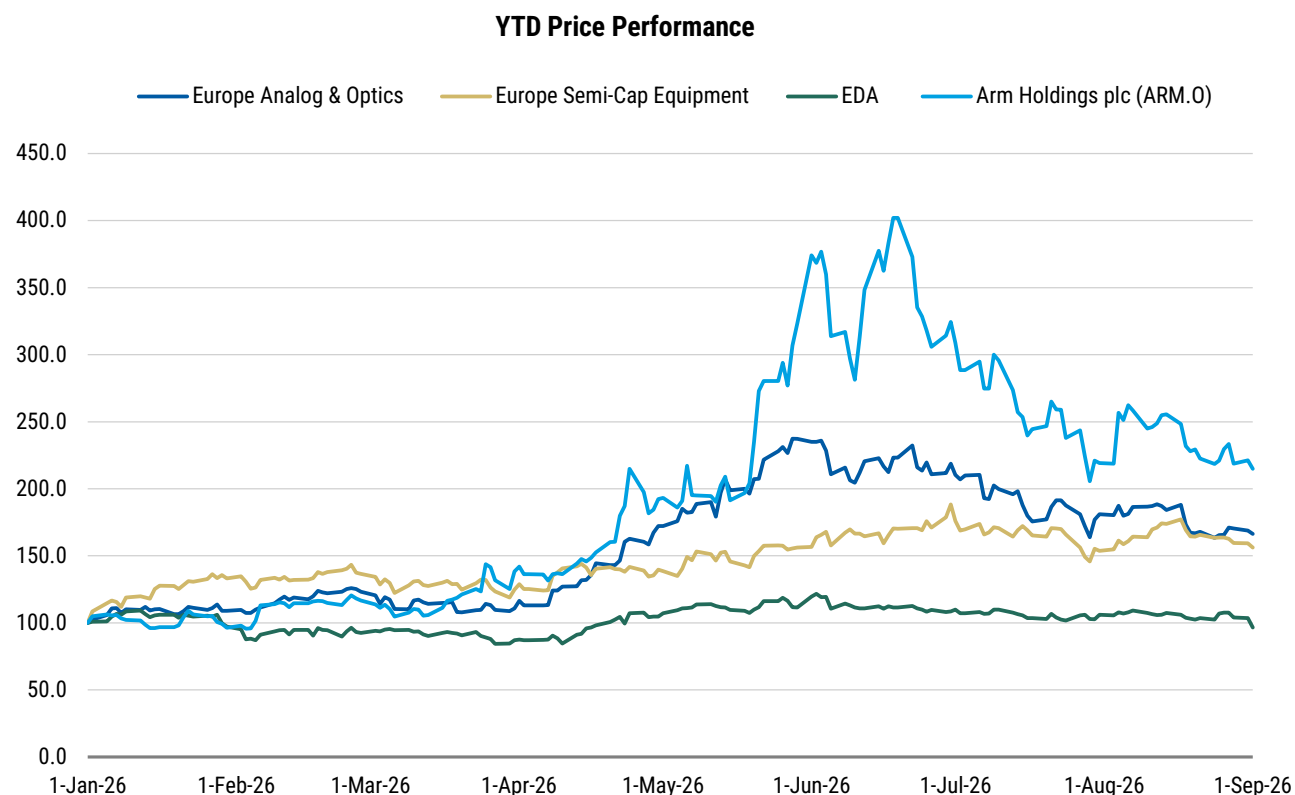


Exhibit 8:

Europe semi-cap equipment outperformed our coverage in 1Q26 while Europe analog/optics and CPUs outperformed in 2Q26.

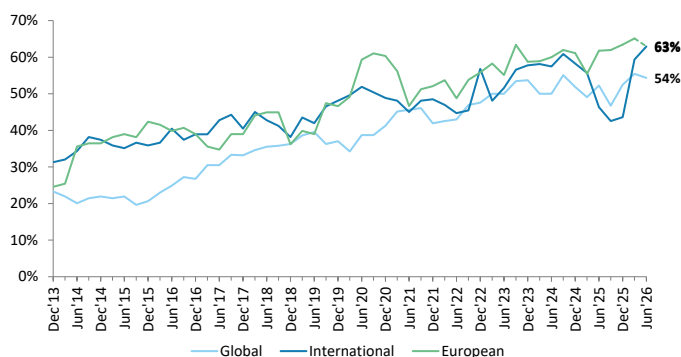


Source: FactSet, Morgan Stanley Research

A pullback in positioning, but the sector remains well owned. Last month, our European Strategists published their latest real money positioning data ([here](#)). While semis remain one of the most-owned sectors, we note a pullback in ASML positioning among both global and European funds. We think this suggests investor interest remains high, with the recent sell-off providing an attractive entry point into our stocks.

Exhibit 9: Fund positioning in European Semiconductors continued to trend up this year.

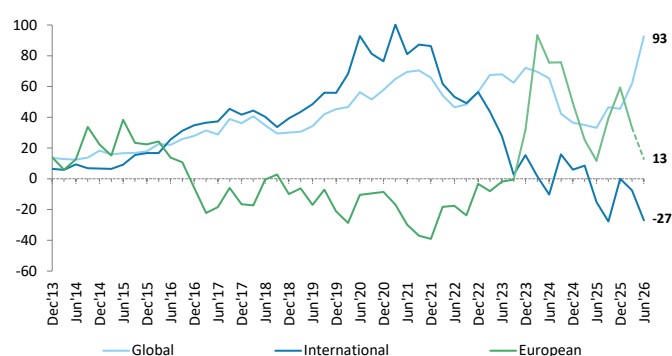
Fund positioning in European Semis & Semis Equip.: % of Funds OW



Source: Note: Our sample group of funds varies month-to-month so is not directly comparable though there is a c. 90% overlap in funds; June data for European funds is preliminary; Source: FactSet, Morgan Stanley Research

Exhibit 10: ASML relative positioning rose sharply for Global funds in Q2 while International and Europe funds trimmed further.

Average Fund Positioning in ASML: Relative Positioning (bp OW/UW)

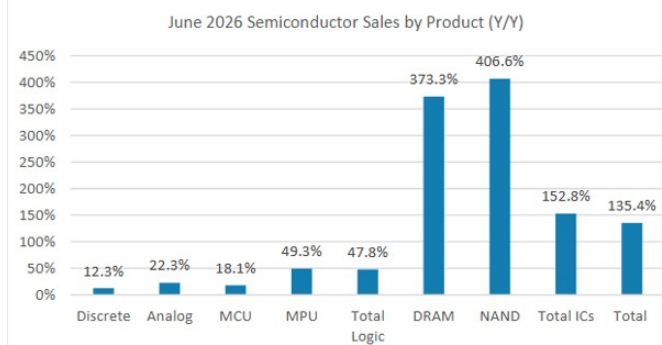


Note: Our sample group of funds varies month-to-month so is not directly comparable though there is a c. 90% overlap in funds; June data for European funds is preliminary; Source: FactSet, Morgan Stanley Research

Wrinkles in the backdrop

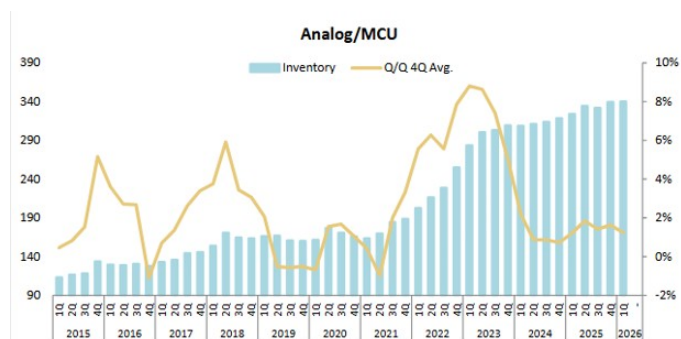
SIA data trends positively... In 1H26 SIA data came in stronger than expected, driven by both broad markets and memory ([here](#)). We think this points to a broadening recovery, with firmer pricing across the supply chain as well as improving demand dynamics off cyclical lows. This supports commentary across our coverage and reinforces our analog thesis that recovery should continue into the latter half of this year. We expect pricing to remain firm, alongside steps towards inventory replenishment and constructive book-to-bills.

Exhibit 11: June 2026 Semiconductor Sales by Product (Y/Y)



Source: SIA, Morgan Stanley Research

Exhibit 12: Inventories remain historically elevated vs history, but momentum has stabilised



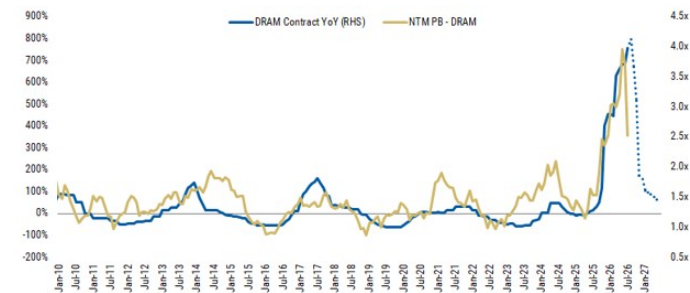
Source: Factset, Morgan Stanley Research

...but DRAM is set to reach an inflection point. The memory-cycle peak is measured by the second derivative of pricing (the YoY pricing peak). DRAM prices are up >600% YoY and look set to reach c.700% YoY in Q3. While our memory team's latest checks suggest no significant change to overall DRAM supply-demand dynamics, for memory prices to continue increasing YoY (%) we would need to see a sizeable c.70% sequential increase in Q4. Consequently, our memory analysts see the cycle transitioning to late cycle by 4Q26, which we expect to weigh on sentiment toward our semi-cap equipment stocks most exposed to memory dynamics. However, we note that structural tailwinds such as Agentic

AI may provide some cushioning against this headwind.

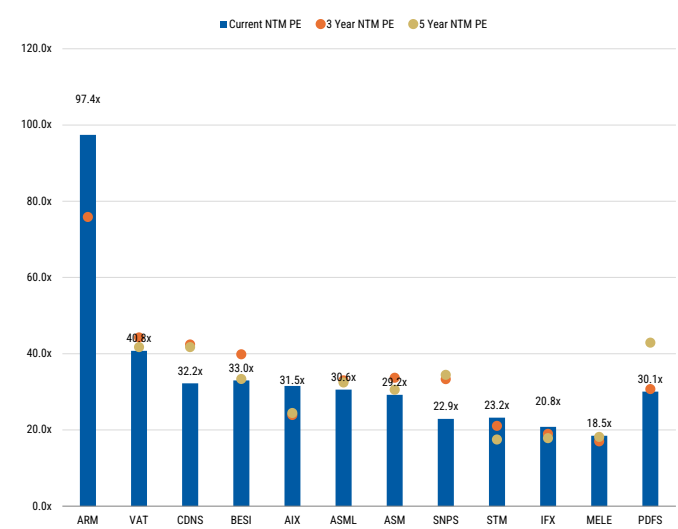
Valuation disparity across our coverage. The greatest disparity versus historical multiples sits within our EDA coverage, where LLM risk to EDA companies has weighed on the stock performance YTD of SNPS and CDNS. For instance, CDNS currently trades at 37x NTM P/E, a discount to the historical 42x 3-year and 5-year NTM averages. Our semi-cap coverage also trades at a discount to historical averages: ASML at 32x (vs. 33x), ASM at 30x (vs. 34x/31x 3-year/5-year), and Besi at 34x (vs. 40x 3-year average). While we think such levels provide attractive entry points, particularly given improving structural tailwinds, we remain selective.

Exhibit 13: DRAM contract YoY vs Memory NTM P/B



Source: Trendforce, Morgan Stanley Research

Exhibit 14: NTM PEs for our coverage



Source: FactSet, Morgan Stanley Research

Balancing Structural Upside with Emerging Risks

ASML and Besi trim PTs, Infineon downgrade to EW and Synopsys upgrade to OW. We outline 4 risks into year end that cause us to shift our relative preferences within the sector:

- (i) GW slowdown risk
- (ii) macro headwinds
- (iii) ARR of LLMs not keeping up with market expectations
- (iv) potential 800v delays.

4 risks from here into year-end

1. GW build-out slowdown risk

While we agree the data center build-out is significant and incremental GW capacity is coming online over the next few years, we see risk that the pace of GW build-out does not keep up with market expectations, and potentially slows. We see two primary risks: (i) grid connection delays, and (ii) political pushback. On the former, we see grid connections as a key bottleneck for data center build-out, with Stephen Byrd, our Global Head of Thematic Research, citing 5–7-year wait times in some regions ([here](#)). Political pushback toward data centers has also grown this year, with concerns centered on three pillars: (i) power bills, (ii) environmental concerns, and (iii) local quality of life. Our Thematic Strategists note that over 300 AI data center moratoriums have been passed in the US since 2023, and we flag this as a potential risk to continued increases in build-out.

2. Challenging macro setup likely to weigh on enthusiasm

Over the last year, NVIDIA has invested in several partners across the semiconductor ecosystem (SNPS, NOK, NBIS, COHR, LITE, MRVL, CRVV), and most recently MediaTek. Our investor conversations have flagged the circularity of these investments as a concern for the broader ecosystem. While our US Semis team does not share this concern, and instead sees NVIDIA's new neocloud structure as driving a new recurring revenue segment ([link](#)), we expect the market to remain focused on these investments into 2H this year. We would also flag the broader macro backdrop as amplifying risk, with UST 30Y yields reaching their highest level since 2007 and oil prices back in focus. Although we continue to like our coverage into year-end, we see risk that this weighs on market sentiment.

Exhibit 15: UST 30Y yields reached highest level since 2007 in August.



Source: Bloomberg, Morgan Stanley Research

Exhibit 16: Our Commodity Strategists recently raised their 4Q Brent forecast to \$100/b.

Dated Brent price forecasts

(\$/bbl)	3Q26	4Q26	1Q27	2Q27	3Q27	4Q27
New	90.0	100.0	95.0	90.0	80.0	75.0
Previous	75.0	75.0	75.0	75.0	70.0	70.0
Change	15.0	25.0	20.0	15.0	10.0	5.0

Source: Morgan Stanley Research

3. Will declining token prices be sufficient to support LLM ROICs?

This summer, broader market focus has shifted toward monthly ARR performance for LLMs. While token costs are declining, we expect focus to shift to whether the acceleration in token price declines is sufficient to support a ROIC narrative for LLMs. Our US Internet team has written extensively on the paths to 25–50% GenAI ROIC for hyperscalers (see [here](#)), citing **token pricing** and **token efficiency** as critical to scaling profitability. Into 2027, we see similar debates coming to the forefront for LLMs.

4. What does a Kyber cancellation mean for 800v native architecture?

Earlier this year, Jensen Huang confirmed that the Kyber form factor had been replaced by another form factor (see [here](#)). While the new form factor has yet to be confirmed, we think the bigger outstanding question is the timeline for 800V-native architecture and whether it has been delayed. We think a delay to 800V-native adoption would pose a risk to companies in the semi ecosystem that deploy 1 or 2 components that may only be used in an 800V architecture. This could have a more significant impact on GaN suppliers, for instance, as we see GaN content increasing materially in the transition from 400V to 800V. To be clear, we aren't saying that 800V-native architecture isn't coming; rather, we are flagging the potential risk.

4 Stock Changes into Year End

We take stock of these risks as well as the 3 key debates per sub-sector that we outline in this note. As such, we trim our PTs on ASML and Besi to €1700 and €220, respectively, but remain OW on both, move Infineon to EW and cut our PT to €65 and move Synopsys to OW, maintaining our \$500 PT.

Exhibit 17: 4 factors underpinning our change in PTs/ratings:

ASML (OW, PT €1700) Stay OW PT €1930 to €1700	BESI (OW, PT €220) Stay OW PT €260 to €220	IFX (EW, PT €65) OW to EW PT €81 to 65	SNPS (OW, PT \$500) EW to OW PT \$500
1. Export restriction risk (e.g. the MATCH Act) continues to be a concern for investors	1. 2Q earnings lacked order visibility for HBM, AMAT deflected a question about its hybrid-bonding outlook, rumours of a de-spec for NVIDIA's Rubin Ultra emerged	1. Risk of 800v delays pushing out data center earnings power and GaN market share	1. Physical AI supports SNPS' simulation portfolio
2. China systems sales development risk	2. Broader memory hierarchies reduce the need for taller HBM stacks	2. Broader VPD competitive landscape	2. Undemanding valuation
3. Capacity concerns remain an overhang for the stock	3. Cut 27e/28e HB tool shipments to 115/190, partly offset by mainstream bonders	3. Risk for weak automotive recovery and MCU pricing environment	3. Factory 2 IP model supportive of Design IP recovery
4. Margin dilution risk from High NA tools	4. We now forecast earnings power of €4.33/€7.09/€7.93 over '26e-'28e, a change of +1%/-4%/-6%.	4. Manufacturing efficiencies from module 4 not present until utilisation at scale in 2028+	4. Greater visibility on profitability from the integration of Ansys

Source: Morgan Stanley Research

ASML: Stay OW, but trim PT to €1700

We remain Overweight ASML as we continue to see DRAM and logic demand as supportive of the margin expansion story. In particular, we think product-led pricing, immersion demand, and capacity upside all support the medium-term setup.

However, there are 4 overhangs into year end. We see 4 key points weighing on near-term sentiment: (i) potential export restrictions, (ii) China system sales development, (iii) capacity concerns and (iv) margin dilution risk.

Risk to China sales has weighed on the stock over the summer with investors focused on the potential implementation of the MATCH Act and the risk of China developing their own EUV tools. While we see limited near-term risk to our estimates, we acknowledge risk to the longer-term outlook.

Capacity concerns have also been an overhang since the Q1 print, with many investors lacking conviction that ASML can deliver more than 110 EUV tool systems in FY28. We disagree, and see commentary from our NDR ([link](#)) as confirmation of capacity optionality. However, we think it will take time for the company to build investor confidence in this expansion. We expect both China sales risks and capacity concerns to remain heavily debated through year-end (see more detail on these debates here: [Semiconductor Equipment](#)).

Margin expansion or margin dilution? Although we think ASML is a margin expansion story, we expect some bears to float the risk of margin dilution from High NA tool

deployment. We disagree that this will be an imminent margin headwind, and instead find the pricing power debate more compelling. However, we acknowledge this debate could weigh on near-term sentiment.

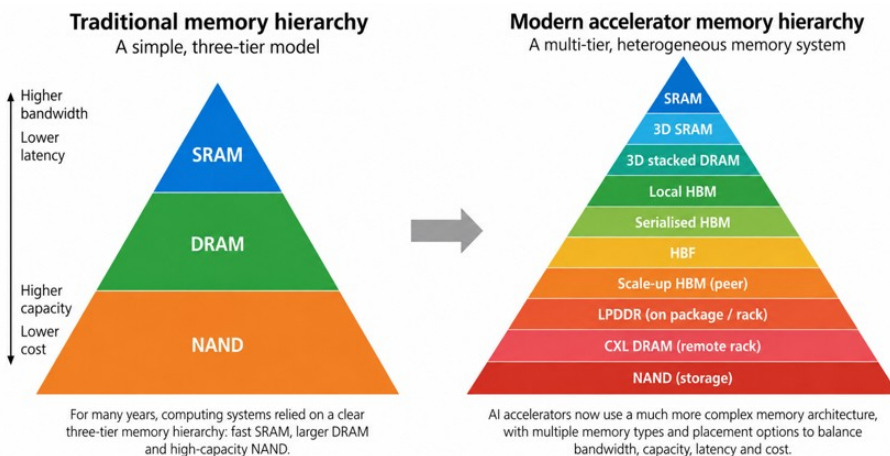
We trim our PT to €1700 (prior €1930). We leave our estimates unchanged but now move to valuing the stock on a 30x P/E multiple, vs 35x previously, at the lower end of ASML's peak cycle multiple range to reflect the headwind to sentiment. Applying a 30x P/E multiple to our FY28 EPS_e of €56.42 moves our PT to €1700.

Besi: Stay OW, but trim PT to €220

June optimism around HBM hybrid bonding has unwound. Besi's shares have sold off since the June investor day, which coincided with peak positive sentiment around the possible adoption of hybrid bonding (HB) for high-bandwidth memory (HBM). Since then, sentiment has soured following a series of events. Besi's 2Q earnings lacked order visibility for HBM, AMAT deflected a question about its hybrid-bonding outlook, rumours of a de-spec for NVIDIA's Rubin Ultra emerged (see our Asia semis checks [here](#)) and, more recently, there was negative commentary from SK Hynix on the readiness of HB even for HBM4e.

Broader memory hierarchies reduce the need for taller HBM stacks. Although hybrid bonding still features on the roadmap for future HBM versions at all three memory makers, the need for hybrid bonding in HBM now appears lower and later than previously assumed. Accelerator architectures presented at the recent Hot Chips conference imply that chipmakers can add seemingly enough capacity through 8- and 12-high stacks, while supplementing this with a much broader memory hierarchy, as illustrated below. This includes stacking SRAM directly on top of accelerators, utilising serialised HBM, peer HBM (within an expanded scale-up domain), LPDDR and CXL. That doesn't mean that adding more HBM next to the package using 16- or 20-high stacks is not useful, especially for architectures that prefer to keep more HBM local (like OpenAI's Jalapeno), but taller stacks are no longer the only way forward.

Exhibit 18: AI uses an extended Memory hierarchy that provides options away from higher stacked HBM



Source: Morgan Stanley Equity Research

We cut 27e/28e HB tool shipments to 115/190, partly offset by mainstream bonders.

For Besi, this reduces the probability of meaningful HBM hybrid-bonder revenue in 27e/28e, and we lower our forecasts accordingly. We now forecast HB tool shipments of 65/115/190 for the '26e-28e period, down from our previous forecast of 65/140/230. This is partly offset by higher revenue in our mainstream-business forecasts, assuming a stronger-for-longer cycle for bonders used for 2.5D (CoWoS-like) applications. As a result, we now forecast earnings power of €4.33/€7.09/€7.93 over '26e-28e, a change of +1%/-4%/-6%.

Our PT moves to €220 from €260 but we stay OW. We cut our PT to €220 from €260, reflecting a combination of lower EPS estimates and a lower FY27e P/E multiple of 31x, down from 35x previously. We stay OW as a lot of this seems priced in and the risk/reward is more balanced, with upside optionality from a stronger mainstream cycle and hybrid-bonding adoption elsewhere (China and wearables).

Infinion Technologies: Move to EW, PT to €65

Structural breakout playing out; limited upside to the narrative for now. While we think Infineon will be the clear data center beneficiary, we see limited upside to the narrative for now. The opportunity appears to be well understood by the market, and we see it unlikely for the company to deliver a Q4 guidance raise that appeases the bulls. We maintain our expectation for €2.8bn data center sales for FY27 and don't expect the data center guidance to be raised to the €4bn+ number some investors are hoping for. This is feeding into street estimates for Power and Sensor Systems (PSS) sales where we now sit 18% and 24% below consensus for FY27 and FY28 respectively (€6.5bn FY27 and €7.9bn FY28).

Emerging risks may prevent the market from stepping in. We see 4 potential risks that could hinder sentiment and Infineon's earnings profile: (i) 800V delays, (ii) VPD competition, (iii) a weaker-than-expected automotive recovery, and (iv) Module 4 manufacturing efficiencies not emerging until 2028. While we leave our estimates unchanged, we think these factors will weigh on near-term sentiment. 800V delays could push out Infineon's power semi revenue, VPD competition could weaken market share, a weaker-than-expected automotive recovery could limit MCU pricing power, and Module 4 manufacturing efficiencies not emerging until 2028 could mean investors' margin expectations have run too far too soon. We are not changing our estimates for now, as we do not see evidence of these risks emerging; however, we flag them as potential downside risks that we think will prevent the market from stepping in.

We trim our PT to €65 (prior €81). We maintain our sum-of-the-parts valuation framework as we agree Infineon is a structural breakout story in the data center. However, we trim our multiple on the server rack business to 30x (prior 45x) as we think this better reflects the broader competitive landscape in the data center and the corresponding growth profile for Infineon. We maintain our 20x P/E multiple for the rest of the group as we continue to view Infineon as moving through a cycle recovery. Applying a 30x P/E multiple to the server rack group and a 20x to the rest of the group on our unchanged estimates sets our PT at €65.

Synopsys: Move to OW, PT \$500

Transition stage complete... Earlier this year we moved Synopsys to EW pending: (i) greater visibility on profitability from the integration of Ansys, (ii) a return to teens %

growth in EDA and (iii) evidence that the co-design offering is as strategically important as competing platforms. We have since gained greater confidence in the synergy trajectory from the Ansys integration and in Design IP recovery. While we are yet to see a return to teens% EDA growth, we think the transition to physical AI will be supportive of this.

...now in prime position for physical AI. We think the movement of AI from a digital/cloud world to a physical world supports Synopsys' portfolio, namely, through the simulation portfolio closing the sim-to-real gap. In our view, this is being overlooked by the market, with few acknowledging the suitability of the company's portfolio for assessing thermal dynamics, mechanical stress and fluid dynamics. We think this is where the Ansys integration comes to the forefront

Could the investor day rebut existing overhangs? We think Synopsys' upcoming investor day offers management an opportunity to signal Design IP recovery, highlight Ansys synergies and challenge market concerns on LLM risk to EDA that we think have weighed on the stock YTD. On Design IP recovery, following discussion at the last print (see [here](#)) we expect management to focus on the new Factory 2 IP model, a customised IP model aimed at hyperscalers and ASIC makers requiring bespoke solutions. We would be looking for direct commentary on the potential upside that this could contribute to Design IP sales. On Ansys synergies, we expect Synopsys to hone in on their Multiphysics Fusion tool, the first joint solution from Synopsys and Ansys, launched in Q3. We see risk skewed to the upside here, with scope for management to provide updated revenue synergies. On LLM risk to EDA, we think the market is looking for a walk through of the company's defensive moat. While some LLMs have claimed internal EDA capabilities, we think this potentially reflects internal use of EDA tools from Synopsys and Cadence. We see scope for a new monetisation model where Synopsys is paid twice: once for the underlying software tools, and again when hyperscalers take up agents from EDA.

An undemanding valuation creates an attractive entry point. Synopsys has de-rated 12% this year, leaving our PT >20% above the market price. Valuation looks undemanding relative to peers as while we agree Cadence should trade at a premium to Synopsys, we think this gap has now become too wide. Synopsys currently trades on a 28x P/E multiple on this year's earnings vs Cadence on a 37x P/E multiple. We continue to value Synopsys on a 33x P/E multiple at the midpoint of the historical trading range of the stock, 30-35x. Applying this multiple to our FY26 EPSe of \$15.12 sets our PT at \$500.

Exhibit 19: SNPS points of focus into the investor day**3 points of focus into the investor day:****(i) Factory 2 IP Model**

- Factory 2 is a new customised IP model aimed at hyperscalers and ASIC makers requiring bespoke solutions.
- A shift from traditional licensing to licence-plus-royalty for these engagements, which would create revenue upside as customer chips reach production.
- We are looking for commentary of this supporting Design IP revenue growth.

(ii) Multiphysics Fusion

- Multiphysics Fusion platform is the first real Ansys synergy.
- Synopsys launched its first joint Synopsys-Ansys product in Q3, integrating thermal analysis directly into the chip design flow – an increasingly binding constraint in AI silicon as power density rises.
- We are looking for updated Ansys revenue synergies.

(iii) LLM risk to EDA

- We think market concern on LLM risk to EDA remains an overhang for the stock.
- While we don't think a formal announcement with an LLM is needed, there does need to be a walk-through of the company's defensive moat.

Source: Morgan Stanley Research

Semiconductor Equipment

Debates:

- Does ASML have the capacity to support WFE demand into 2028?
- Are export restrictions a risk for Semi-Cap?
- Can WFE growth continue to accelerate in 2028?

Market View: ASML continues to be viewed as a bottleneck for WFE expansion with the market unconvinced on the buildout of the Brainport Industries Campus (BIC). Market concern on export restriction implementation also appears significant.

Our View: We view both EUV capacity concerns and export restriction concerns as overdone. While we acknowledge both as risk to the downside for our estimates, we think the likelihood of this materialising is low. Instead we think the WFE growth trajectory into 2028 is the more pressing issue.

1. Does ASML have the capacity to support WFE demand into 2028?

EUV capacity concerns have weighed on ASML shares since the start of the year. While management provided incremental detail on capacity at both the 1Q26 and 2Q26 prints, the commentary continued to disappoint investors. At the 1Q26 print, management outlined capacity for at least 60 low-NA EUV systems in 2026 and 80 in 2027. At 2Q26, management subsequently raised the 2027 figure to 85 tools and introduced a 2028 capacity estimate of 110 units. However, the market had been looking for 90 EUV tools in 2027.

In our view, this commentary better resembles a capacity floor than a ceiling. At our NDR, ASML management stressed flexibility around capacity additions, citing significant cuts to EUV assembly time, with efficiency further bolstered by AI-driven diagnostics. Interestingly, management highlighted the Brainport Industries Campus (BIC), which it expects to be operational in 2028. While our understanding is that the initial use case will be for DUV, we think the campus could accommodate EUV if necessary, suggesting potential upside to existing capacity guidance. **For 2027, we think the debate will centre on 1) the Brainport Industries Campus buildout, and 2) whether ASML is a bottleneck for WFE expansion.**

Exhibit 20: ASML outlined at the April AGM plans to start building out the BIC in 3Q26.

Expanding our footprint to serve our customers

Preparing for growth

Asia	US	Europe
 Korea's Hwaseong campus In late 2025, Korea's Hwaseong campus opened its doors, enabling us to collaborate more closely with our key customers and the broader industry ecosystem in South Korea.	 Phoenix training center In Q4 2025, we inaugurated our new training center for customer support engineers in Phoenix, Arizona – a significant milestone in our global initiative to provide technical training support to field teams and customers.	 Eindhoven BIC campus Preparations of our BIC campus are progressing. We plan to develop the campus in phases and have the first employees move to the new location in Q1 2028. Groundbreaking planned in Q3 2026.

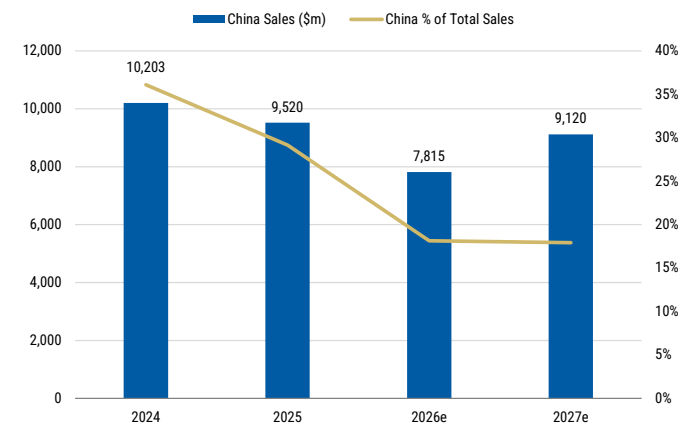
Source: ASML AGM Presentation

2. Are export restrictions a risk for Semi-Cap?

The debate around export restrictions on sales to China has picked up this summer, with the potential implementation of the MATCH Act remaining an overhang for many of our semi-cap equipment names. Drafted in early April 2026, the proposed bill would extend existing controls for ASML to the full immersion DUV range and servicing provisions. As such, it could pose a risk to both system sales and Installed Base Management sales.

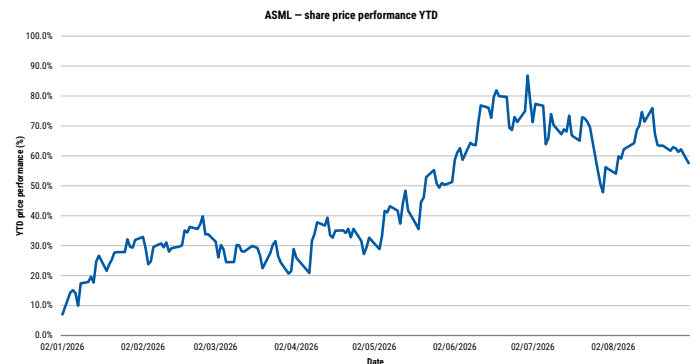
We maintain our view that implementation would not have an immediate impact on ASML sales, although approval of the bill could pose a headwind to our FY27+ estimates. If enacted, governments would have a 150-day window to adopt the export restrictions; therefore, as it stands, we see no risk to our 2026 estimates. We currently expect China to account for 18% of ASML total sales in both FY26 and FY27. We also think risk to the downside is limited for our FY27 estimates as, we currently view implementation as unlikely given (i) the likely Dutch government pushback ([link](#)), and (ii) the broader scope of the MATCH Act meaning that it would impact the entire semi-cap ecosystem, not just ASML.

Exhibit 21: We currently model ASML's China sales declining as a proportion of total sales FY25 to FY26.



Source: Company data, Morgan Stanley Research estimates

Exhibit 22: ASML traded down 10% on market fears of Chinese competition



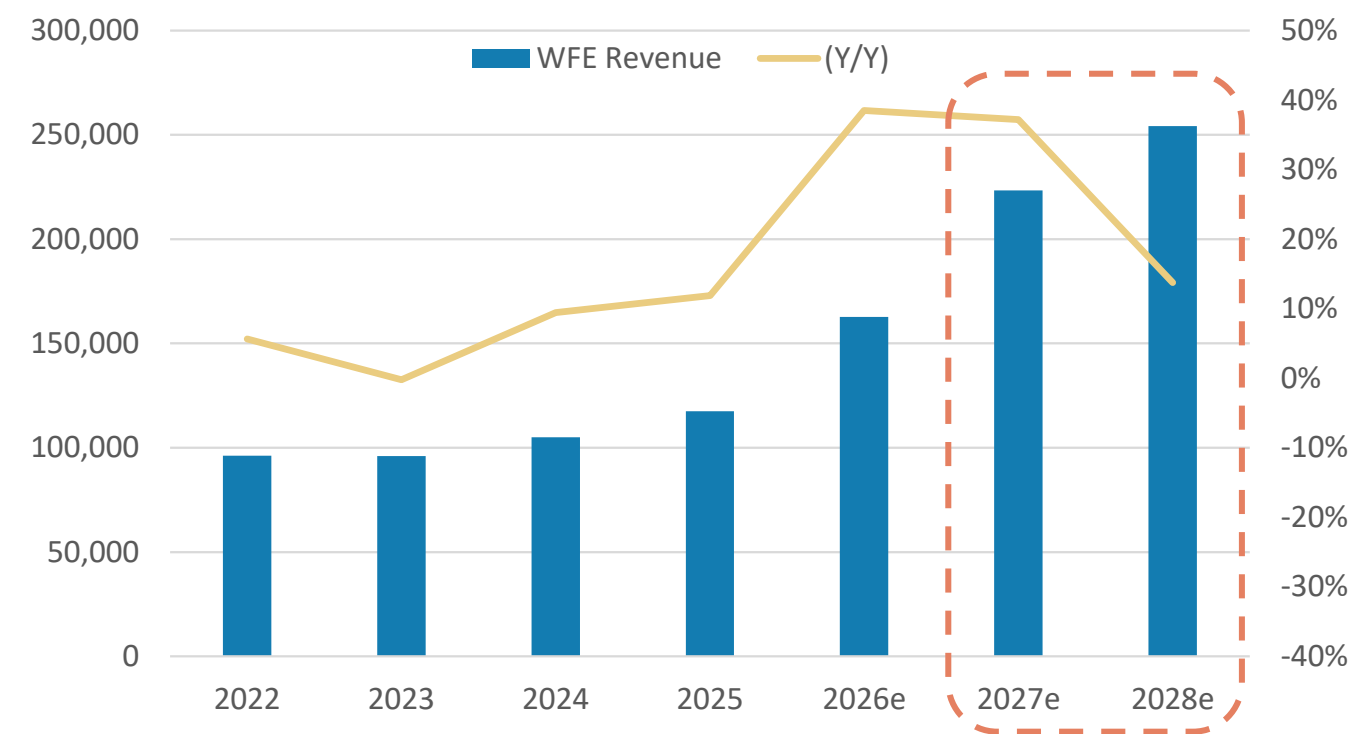
Source: FactSet, Morgan Stanley Research

3. Can WFE growth continue to accelerate in 2028?

WFE growth is accelerating into 2027, but we think this is well understood. We have revised up our WFE forecasts 5 times YTD as strength of demand and the pace of supply additions continue to outstrip expectations (see our most recent update [here](#)). We now forecast 2026 WFE coming in at \$163bn reflecting significant **DRAM** demand as well as greater spending from leading logic suppliers such as TSMC, Samsung and Intel. Looking to 2027, we are also incrementally bullish modelling WFE growing at 37% yy to \$223bn. This reflects (i) Intel becoming a meaningful #2 spender of Logic, (ii) 500kwpn of DRAM greenfield capacity and (iii) a restart in NAND greenfield investment. However, our investor conversations suggest that 2027 being a strong year for WFE is well understood by the market.

We think the debate now centres on 2028 and whether growth will continue to accelerate. We agree 2028 demand will remain strong and model this reaching \$254bn. However, growing only 14% yy implies a meaningful growth deceleration from the 37% yy growth in 2027. For growth to continue to accelerate, WFE would need to exceed \$300bn in 2028, which we don't currently see as likely. **We think the market debate now shifts to 2028 supply/demand dynamics, with investors focused on the 2028 capex outlook from memory makers, leading logic suppliers and hyperscalers.**

Exhibit 23: We see it unlikely for WFE growth to continue to accelerate into 2028.



Source: Morgan Stanley Research estimates (e)

Analog and Optics

Debate:

- Can margin recovery keep pace with market expectations?
- Who are the beneficiaries of the transition to vertical power delivery?
- How does multi-rack scale up increase Soitec's revenue opportunity per GPU?

Market View: Expect margins to ramp near-term and intrigue on players in vertical power delivery.

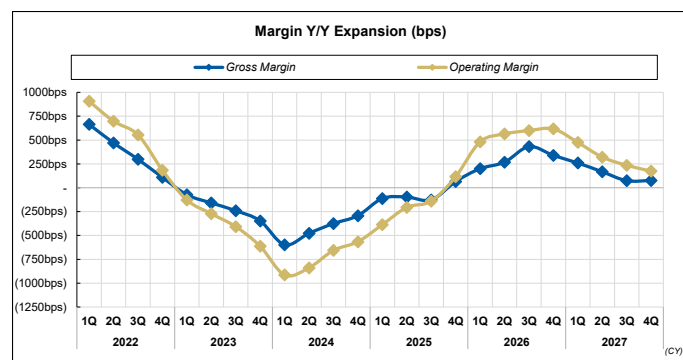
Our View: We exercise caution in modelling significant near-term margin expansion. Instead, we think this is a longer-term development as start-up costs and other expenses associated with the IDM business models weigh on margins. We note more competition in vertical power delivery and see the power efficiency vs power delivery debate coming to the forefront.

1. Can margin recovery keep pace with market expectations?

Q2 prints saw margins miss the mark across our European Analog coverage. While the direction of travel remained positive, the pace of improvement was not enough to clear a bar that had moved higher through the quarter. STMicro was the clearest disappointment. Despite an improving Q2/Q3 gross margin profile (Q3e: c.37% up >200bps qq), the Q4 guide underwhelmed. A mix of proforma charges and ramp-up costs mean the hoped-for 40%+ margin looks unachievable, despite a sales guide reaching the landmark \$4bn+/Q level. Infineon echoed similar margin disappointment, with c.23% Q4 segment-margin guidance falling short of the mid-20s% buy-side expectation, as mix and FX offset the benefits of stronger AI and cyclical volumes

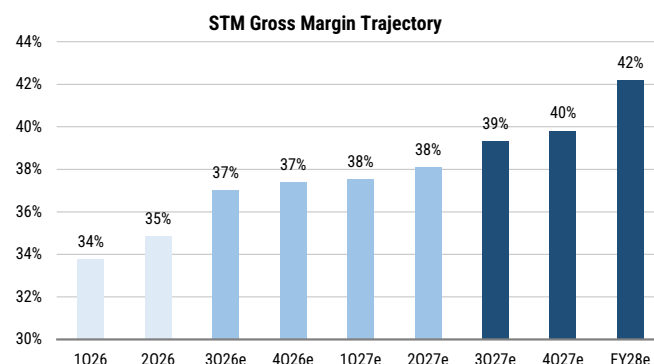
We think margin expansion will continue, but is the market overestimating the pace of this recovery? We agree changes in mix and more favourable supply/demand conditions mean that margins will continue to expand. However, we think the market is failing to acknowledge the IDM business models of STM and IFX versus TXN and ADI. This means additional headwinds that we think the market failed to recognise last quarter such as start-up costs and utilisation charges. For instance we see a gradual margin ramp for the likes of IFX and STM where start-up costs of the Dresden and Crolles fabs continue to weigh on accelerated margin expansion pathway. **We agree margins are improving for both but exercise caution in the ramp that some were expecting at the last print.**

Exhibit 24: We see global analog margins continuing to expand into 2027...



Note: Represents total of defined Core 8 companies (TXN, ADI, IFX, STM, NXPI, Renesas, MCHP, ON)
Source: VisibleAlpha, Morgan Stanley Research estimates.

Exhibit 25: ...consistent with our STMicro margin progression into FY28.

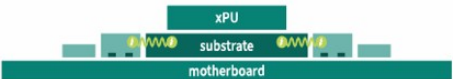
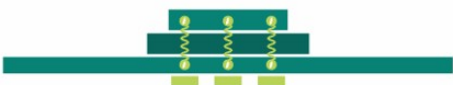


Source: Company data, Morgan Stanley Research estimates (e)

2. Who are the beneficiaries of the transition to Vertical Power Delivery (VPD)?

The evolving competitive landscape in VPD. This year, we have seen plenty of M&A activity across the analog sector focused on acquiring key technology to power the data center. Most recently, the focus has been on Vertical Power Delivery, with global analog companies either acquiring Voltage Regulator Modules (VRMs) or technology surrounding the architecture. Earlier this year, we identified Infineon and Monolithic Power Systems as the key players with VRM technology. However, we note the recent emergence of ADI and Navitas as potential competitors, with the former acquiring Empower Semiconductor ([here](#)) and Navitas announcing plans to acquire Claros ([here](#)). While Infineon has always had VRMs, we think Infineon's recent announcement to acquire C2i Semiconductor ([here](#)), a provider of software-defined multiphase controllers and smart power stages for AI DC applications, is interesting as it further broadens Infineon's data center portfolio.

Exhibit 26: IFX's VPD solutions focus on power efficiency.

 Lateral Power Delivery	Discrete (Lateral) + Power stages, inductors and capacitors located next to the processor + Lowest cost, with established eco-system and quality record + PDN losses exceed 100W for GPU currents beyond 850-1000A	Lumped PDN ¹ 90-140μΩ 	
 Vertical Power Delivery	BVM – Backside Vertical Module (Vertical) + Increases power density by eliminating required spacing between multiple smaller modules + Simplifies motherboard design by eliminating routing of input power and control signals under processor	Lumped PDN ¹ 10-15μΩ -89% 	
 Vertical Power Delivery	SiVR – Substrate integrated Voltage Regulator (Vertical) + Reduces substrate PDN losses by additional 10-15% + Removes substrate interconnect current limitations	Lumped PDN ¹ 7-10μΩ -93% 	

Source: Infineon company presentation.

Power efficiency or power density? Last November, we launched our server rack model, where we estimate power semi content per rack and Infineon's market share by element. In our discussions with investors, the topic of power efficiency vs. power density came up frequently, in particular, how MPWR's and IFX's power architectures differ and which may be the preferred play in the second stage of the data center. While we think both will do well, we believe power efficiency will become increasingly important as data centers scale and will be a characteristic rack providers consider when selecting their preferred provider. As such, we find Infineon's acquisition of C2i Semiconductor particularly interesting, enabling Infineon to maximise power transmission from the VRM to the GPU and, in our view, supporting our thesis that Infineon is the preferred power semi provider in the data center. **We expect this debate to continue into the latter half of this year, with investors focused on the different technology offerings from analog companies and what this implies for respective data center revenue.**

3. How does multi-rack scale-up increase Soitec's revenue opportunity per GPU?

Multi-rack scale up supports NPO/CPO-based optical connectivity... The Vera Rubin POD was introduced during the company's annual GTC conference in March 2026 and is the best-known example of a multi-rack set-up. NVL576 combines eight separate MGX NVL racks, each with 72 Rubin Ultra GPUs, all in a single 576-GPU NVLink domain with copper within the racks (no Soitec content) and direct optical connections between the racks (via either NPO or CPO, both of which are enabled by Soitec's silicon photonics wafers). However, NVIDIA is not the only company that has announced a multi-rack pod: our Asia team reported that AI "supernodes" enabled by scale-up are becoming the focus of China's AI computing landscape with companies like Biren and Huawei demonstrating scale-up domains of 1,024 accelerators ([link](#)). Connecting multiple racks within one scale-up domain increases the need for high-bandwidth links over distances where copper

becomes less practical, supporting NPO/CPO-based optical connectivity.

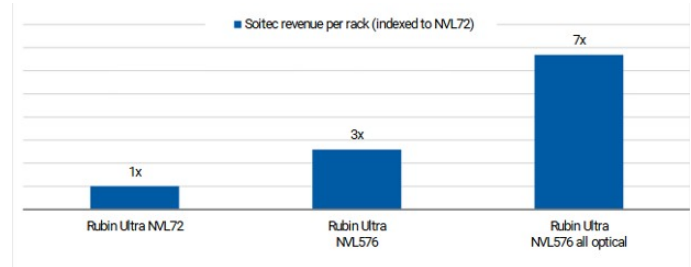
...driving a sizeable revenue increase for Soitec. Soitec said that most of its revenue for the current fiscal year is from pluggables, but also partly from NPO. The clearer timelines (mid '27, mentioned by several ecosystem partners) and the necessity (reducing HBM content per GPU) mean that this is likely a meaningful, relatively near-term catalyst for Soitec. Our calculations suggest that this could triple the company's revenue per GPU relative to a "standard" rack containing 72 GPUs with no direct connections to other racks. We don't expect the market to immediately switch to this type of rack topology, but even assuming a stepwise increase towards 30% over multiple years could see Soitec's revenue per GPU grow by c.50%. The replacement of copper inside a rack remains the larger opportunity (and comes on top of this development), but as timelines remain somewhat uncertain we currently don't explicitly model this out.

Exhibit 27: NVIDIA Polyphex prototype, a fully functional GB200-based multirack NVL576 scale-up system.



Source: NVIDIA

Exhibit 28: Soitec revenue per rack shows a sizeable content increase for multi-rack scale-up topologies.



Source: Morgan Stanley Research estimates

The positive mix impact of photonics is largely missed by the market. We are ahead of the Street on revenue for fiscal '27e-'29e (1%/5%/16%) which is driven by a much more positive view on Soitec's photonics business, offset by some conservatism on Soitec's mobile and automotive divisions. To our surprise this gap widens meaningfully on the gross profit line, here we are 10% and 36% ahead of the Street, despite having seen consensus come up quite materially post earnings (19%/14% for FY27/28). Although there is a natural element of gross margin expansion with higher growth, we also believe that the market is not fully aware of the positive mix that comes with the outperformance of photonics.

EDA/IP & Design

Debate:

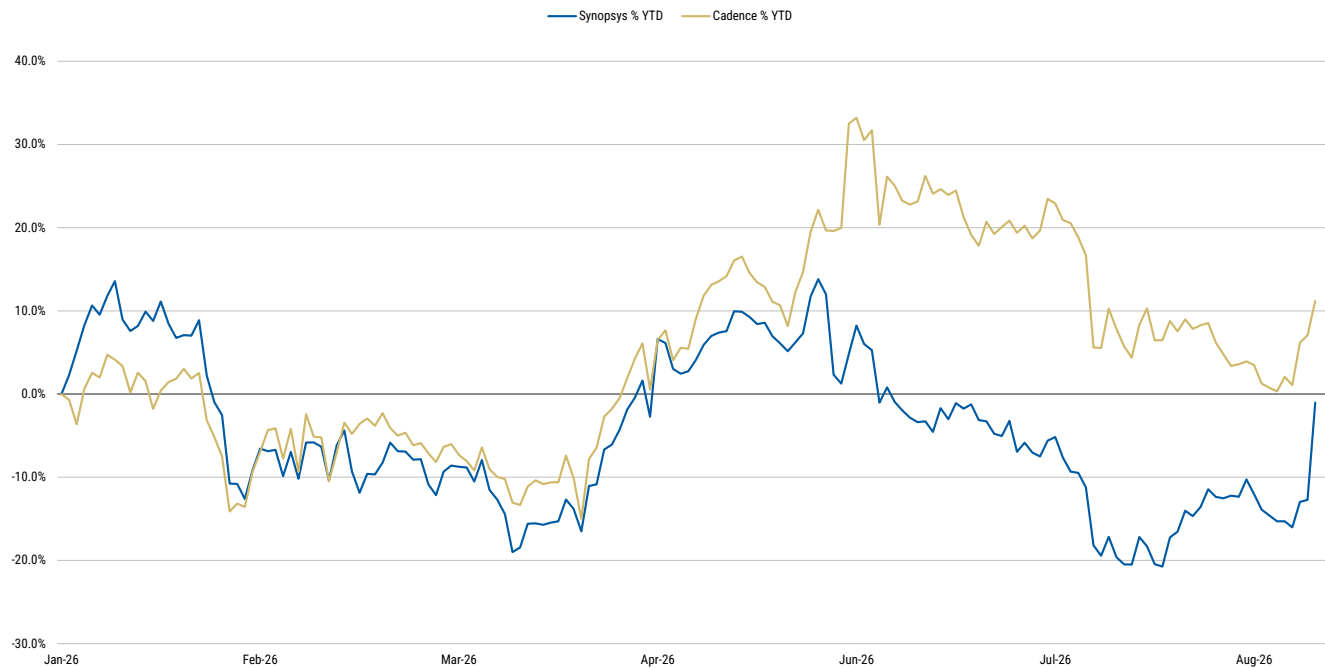
- Are LLMs a credible threat to EDA?
- Is 27 where we will begin to see earnings evidence of simulation and physical AI?
- How large is the CPU TAM?

Market View: EDA sentiment still appears weak but we think interest stems from software sentiment recovery, relative valuation gap (especially for SNPS), and the possibility of simulation technology and where it could drive earnings upside into 2027. Arm, meantime, is viewed as an Agentic CPU winner with eyes on how much of the CPU TAM the company can garner.

Our View: We think concerns are overdone regarding LLM risk to EDA, but don't expect market sentiment to be a quick fix. Instead, proof of earnings resilience is needed, including evidence of momentum in core EDA growth. While we agree Arm's transition to chip-making is a major source of debate.

1. Are LLMs a credible threat to EDA?

LLM risk has weighed on EDA sentiment. Share price action with both Synopsys and Cadence has struggled YTD as concerns around LLM-enabled EDA capabilities have caused market sentiment to falter. The most notable move came after Moonshot AI announced Kimi K3, claiming the LLM could autonomously design a functional semiconductor chip in 48 hours using open-source EDA tools; SNPS and CDNS traded down 10% on the day. Despite management efforts to abate concerns, we think the market, similar to global software stocks, is awaiting proof of earnings resilience before becoming more willing to step in. **We think this is overdone and expect market concerns to alleviate towards year end.** The use of an open source EDA to design a 45nm chiplet is a far cry from high volume, commercial grade production of leading edge AI chips at 2/3nm.

Exhibit 29: SNPS and CDNS have traded down on LLM news flow this year.

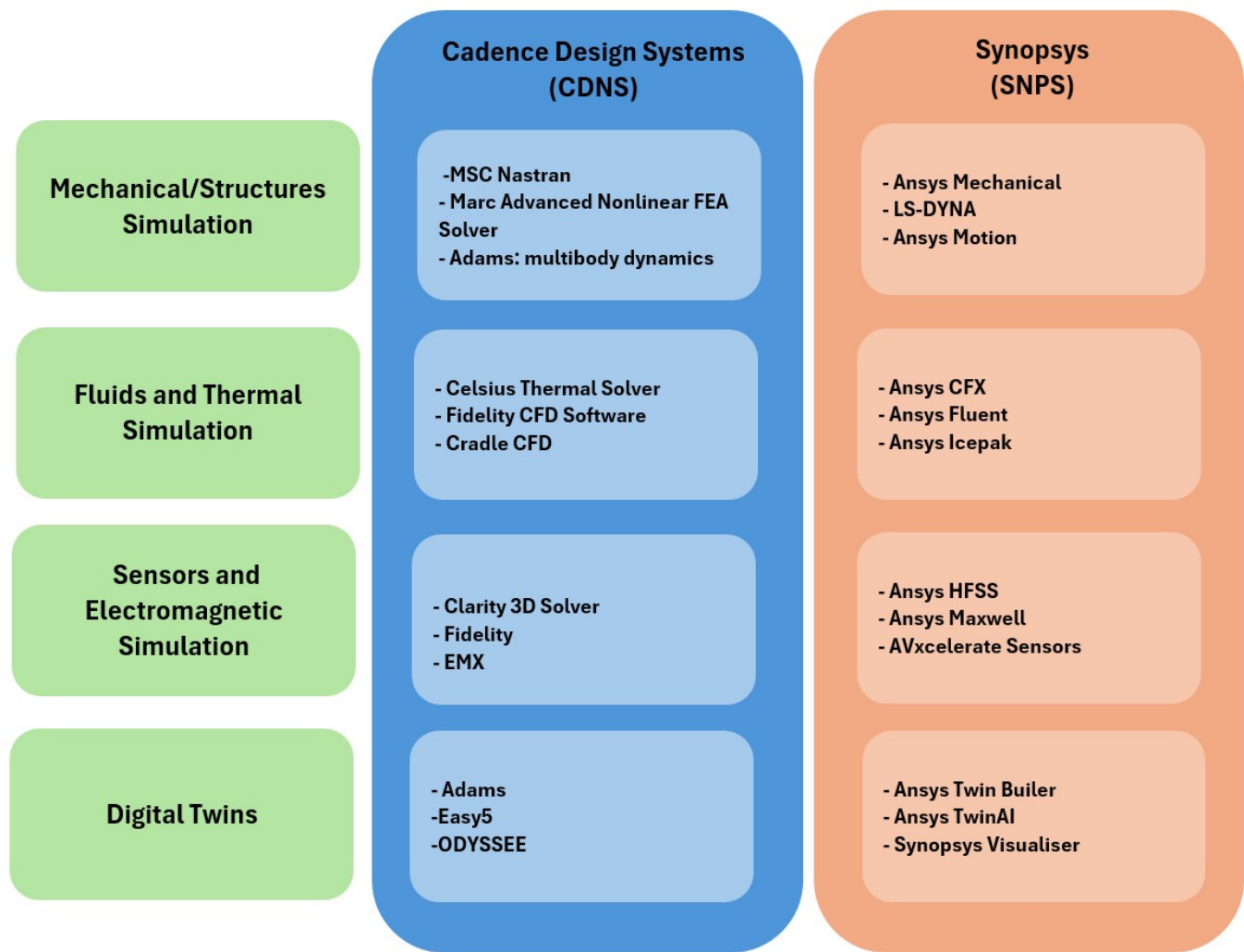
Source: FactSet, Morgan Stanley Research

2. Is 2027 where we will begin to see earnings upside from simulation?

Physical AI is coming, and, in our view, SNPS and CDNS are positioned to benefit.

While LLM risk has been at the forefront of investor debates, we think this overlooks their simulation portfolio. Our Global Embodied AI Strategists have written about autonomous vehicles and humanoid commercialisation being deployed in the near term. We think the movement of AI from a digital/cloud world to a physical world supports Synopsys and Cadence's portfolios, namely, through the simulation portfolio closing the sim-to-real gap. In our view, this is being overlooked by the market, with few acknowledging the suitability of the companies' portfolios for assessing thermal dynamics, mechanical stress and fluid dynamics. In particular **while we don't see Synopsys' Investor Day on the 30th September as a significant positive catalyst for the stock, we do acknowledge that this may create scope for management to circle in on the simulation opportunity.**

Exhibit 30: Simulation technology from Cadence and Synopsys

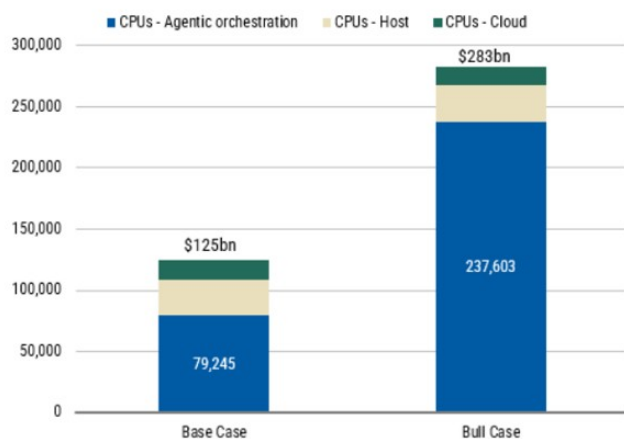


Source: Morgan Stanley Research

3. How large is the CPU TAM and who benefits?

Agentic AI puts the CPU in the spotlight. 1Q26 was an agent AI breakout quarter with a surge in 'agent-first' strategies across many companies, while in 2Q26 focus shifted to the speed of agentic AI integration. We think Agentic AI lifts the overall CPU TAM incrementally higher to US\$125bn by 2030 while our bull case suggests this could reach \$283bn by 2030.

Exhibit 31: We model Agentic AI driving the CPU TAM incrementally higher in 2030.



Source: Morgan Stanley Research estimates

Exhibit 32: CPU vendors share prices have inflected with the rise of the AI agent



Note: CPU companies include: AMD, INTC, ARM. Source: FactSet, Morgan Stanley Research

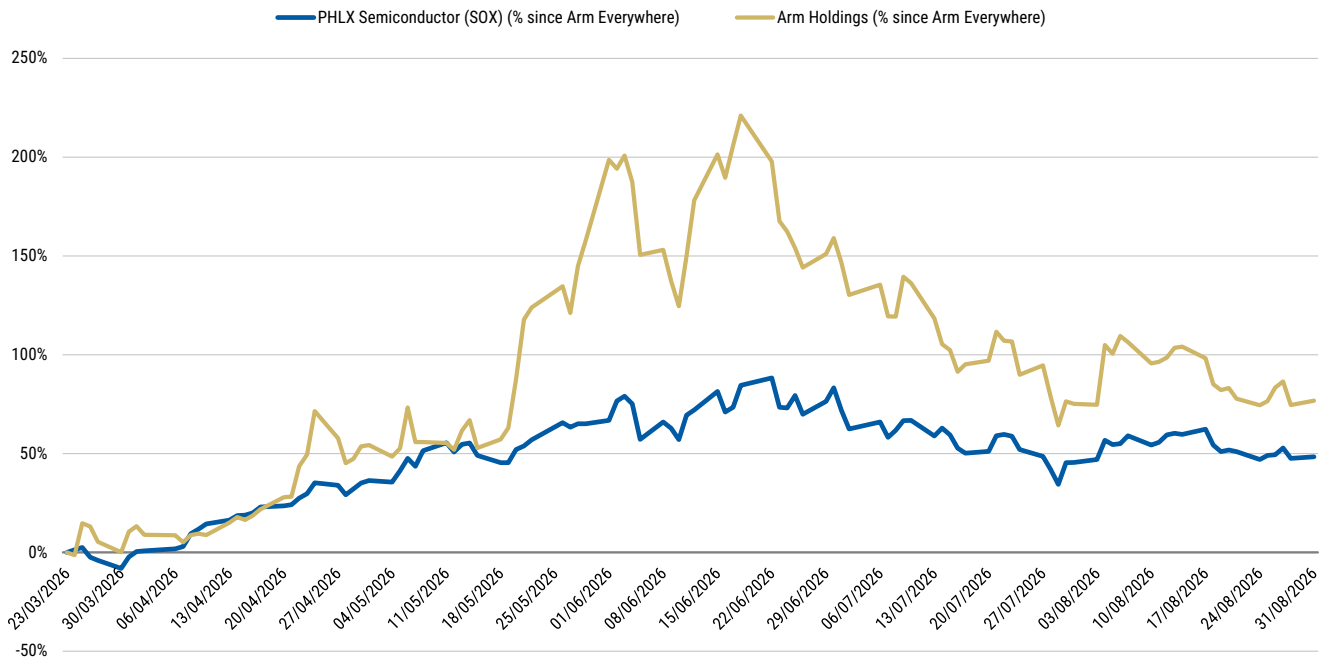
How much of this TAM can Arm capture? At the last print Arm raised their outlook for AGI CPU demand, now expecting more than \$2bn of AGI CPU demand across FY27–28, more than double the opportunity identified at launch. The use case is also broadening beyond AI-accelerator head nodes to include standalone cloud servers and agentic-AI orchestration, where CPUs coordinate agents, memory and inference across clusters. Initial products have been delivered to multiple customers but secured manufacturing capacity currently supports at least \$1bn of cumulative revenue through FY28. **The key investor debate is therefore conversion rather than demand: wafer, substrate, test and memory availability will determine how much of the pipeline can be monetised, and when.**

How we position in our coverage

ARM - EW, PT \$212

Our thesis: further upside dependent on supply-backed CPU conversion and sustained royalty growth. Arm's strategic transition from IP licensing towards merchant silicon materially expands the long-term opportunity, but also raises execution risk, capital intensity and potential channel conflict. The shares have derated from recent highs but still trade on a demanding multiple, while the limited free float amplifies volatility. We remain Equal-weight with a \$212 price target: the stock already discounts a substantial portion of the 2030 CPU opportunity, leaving further upside increasingly dependent on supply-backed AGI CPU conversion and sustained data center royalty growth.

Exhibit 33: Arm shares have de-rated since the June peak but still remain elevated vs the SOX.



Source: FactSet, Morgan Stanley Research

Catalysts: The principal near-term debate centres on royalties mix. In our view, smartphone royalties remain exposed to weaker mid-tier Android demand, while data center royalties provide the cleaner structural offset. We think evidence of improving smartphone builds would help alleviate market concerns on handset royalty weakness. While longer term we see the market focused on the CPU opportunity, the earnings potential as well as potential margin headwinds, we also think a v10 architecture could be coming which could pique market intrigue. Our sense is that this would be an agentic architecture designed to support the move of AI to the edge.

Valuation: retain Equal-weight and \$212 price target. We maintain our c.\$125bn 2030 CPU TAM and unchanged revenue assumptions. Our sum-of-the-parts valuation applies 50x EV/EBIT to the IP business and 20x to the emerging chip business. We view data

center royalties as the most credible source of near-term earnings upside, but believe the current valuation already captures a large part of the longer-term CPU opportunity. Upside requires faster supply expansion, stronger AGI CPU conversion and sustained royalty-rate progression; downside centres on slower silicon execution, elevated operating costs, lower chip margins and potential licensee conflict.

ASMI - OW, PT €1100

Our thesis: main beneficiary of stronger and broader leading edge logic/foundry capex.

Leading edge foundry capex is inflecting sharply higher, which should increase supply in the face of heavy AI demand. ASMI has highest exposure to this and should see continued share of wallet gains off every capex dollar spent. We see strong growth for the years ahead.

Catalysts: As a beneficiary of logic demand, growing WFE demand will support ASMI. WFE growth is accelerating into 2027, but we think this is well understood. Instead, we think market debate now centres on 2028 WFE and whether growth will continue to accelerate.

Valuation: retain Overweight rating and €1100 price target. ASMI currently trades at a 5 turns discount to the semi-cap group despite having one of the strongest (and durable) growth profiles in the sector. We are OW with a €1,100 PT, as we apply a 33x P/E multiple to FY27 Earnings.

ASML - OW, PT €1700 (Top Pick)

Our thesis: DRAM and Logic demand accompanied by margin expansion. ASML is a dominant supplier of critical toolsets for the semi manufacturing process. We anticipate consensus earnings revisions as debate shifts to 2027 on improving DRAM and logic demand for EUV. We expect margin expansion over the mid-to-long term as ASML can apply value-based pricing as a medium-term lever.

Catalysts: As outlined earlier in the note we see 4 key points weighing on near-term sentiment: (i) potential export restrictions, (ii) China system sales development, (iii) capacity concerns and (iv) margin dilution risk. We think that evidence against these overhangs will improve sentiment and support a rerating. In addition, new entrants such as Terafab continue to generate intrigue amongst investors, and could add earnings upside.

Valuation: retain Overweight and lower PT to €1700. We value ASML at a P/E multiple of 30x, at the lower end of our usual cycle peak valuation range of 30-35x. This is in line with historical peaks, but we note upside potential from significant foundry capex and rising DRAM prices. Applying a 30x P/E multiple to our €56.42 FY28 EPSe gives us a PT €1,700.

BE Semiconductor Industries - OW, PT €220

Our thesis: Mainstream to stay stronger for longer, but HBM likely to come later and lower. The set-up for Besi's core business continues to improve, evidenced by a need for chiplet architectures (2.5D) for AI accelerators. Accelerating hybrid bonding adoption in AI

(both compute and networking) has started to result in meaningful orders.

Catalysts: We stay OW as a lot of the hybrid bonding risks seem to be priced in and the risk/reward remains attractive, with upside optionality from a stronger mainstream cycle and hybrid-bonding adoption elsewhere (China and wearables).

Valuation: We apply a 31x multiple to FY27 earnings power which is at the lower end of the company's 5 year average P/E multiple range of c.30-35x and reflects the emergence of risks to HB adoption. Valuation in the high 20s FY27e P/E is not reflecting optionality of broadening hybrid bonding adoption outside of HBM.

Cadence Design Systems - OW, PT \$400

Our thesis: An agentic and physical AI beneficiary. We think the range of growth drivers and the operational efficiency inherent in Cadence's business model has yet to be fully reflected in the share price. The company has a solid base in systems analysis and simulation work in the chip design flow, deep customer relationships, a leading position in analog and mixed-signal design, a solid balance sheet and should enjoy strong structural growth with its new AI-driven EDA tools.

Catalysts: In our view, agentic EDA increasingly looks like an additive, durable growth vector rather than a near-term hype cycle. We think monetisation of this full agentic suite remains in focus into the latter half of this year with eyes on potential revenue upside into 2027. We think near-term expectations are grounded, as guidance doesn't assume a sudden step-function in adoption. Therefore, we see room for management to surprise to the upside with this commentary. In addition, with growing evidence of physical AI into next year, we expect investors to begin to take note of Cadence's simulation portfolio, which we think remains underappreciated by the market.

Valuation: retain Overweight and \$400 PT. We apply a c.41x P/E multiple to our \$9.71 FY27e EPS, an upcycle P/E within the 40-50x historical P/E multiple range. Our view remains that Cadence deserves a premium to peers, given its position at the centre of the semis/systems convergence and it is likely to benefit from several growth drivers across FY26-27e.

Infineon - EW, PT €65

Our thesis: Structural breakout story playing out, but priced in. The structural breakout story is playing out and although we think Infineon will be the clear data center beneficiary, we see limited upside to the narrative for now. The opportunity appears to be well understood by the market and we don't think the company can deliver a guidance raise that will appease the bulls. We maintain our expectation for €2.8bn data center sales for FY27 and don't expect the data center guidance to be raised to the €4bn+ number some are hoping for.

Catalysts: For data center power updates we now look to the AI Infra Summit in September and the OCP Global Summit in October as opportunities for incremental commentary on the timeline for 800v rack deployments and adoption of power semi components.

Valuation: move to EW, lower PT to €65. We maintain our sum-of-the-parts valuation

framework as we agree Infineon is a structural breakout story in the data center. However, we trim our multiple on the server rack business to 30x (prior 45x) as we think this better reflects the broader competitive landscape in the data center and the corresponding growth profile for Infineon. We maintain our 20x P/E multiple for the rest of the group as we continue to view Infineon as moving through a cycle recovery. Applying a 30x P/E multiple to the server rack group and a 20x to the rest of the group on our unchanged estimates sets our PT at €65.

Melexis - EW, PT €65

Our thesis: Underperforming the peer group in FY26. Melexis enters 2H26 with higher revenue expectations, but growth still trails peers despite a weaker first-half base. Margin conversion remains muted as management prioritises volume over pricing and wage inflation absorbs gross margin gains, while robotics remains too small to affect the near-term earnings path. We remain Equal-weight pending clearer evidence of sustained catch-up in growth and stronger operating leverage.

Catalysts: We think as physical AI becomes more evident and humanoid deployment picks up pace, we could see growing earnings power from Melexis' sensors. Robotics is becoming more tangible, but remains an outer-year item. Management frames 2025 and 2026 as funnel-building years, with revenue growth in 2027/28 and a more visible contribution after 2028. We think evidence of this at the next prints would be upside to the story.

Valuation: remain EW, PT €65. We apply a c.18x P/E to FY27 earnings to derive our €65 price target. 18x is slightly below the analog/automotive peer group but fair given the absence of AI exposure.

Nordic Semiconductor - EW, PT 185 NOK

Our thesis: Strong momentum reflected in valuation. Nordic Semiconductor is well placed to capitalise on robust IoT market growth and a favourable product cycle driven by the nRF54 platform, but we remain Equal-weight as these positive trends appear priced in at current levels. The IoT cycle and nRF54 should carry growth, but Nordic has limited direct exposure to the AI spending driving much of the sector, and we currently lack conviction to underwrite sustained greater than 20% growth, seeing better opportunities elsewhere.

Catalysts: Sustained momentum is critical for the stock, making upcoming quarterly reports pivotal for its trajectory. Beyond that, tangible signs of nRF54 adoption by customers would be a positive.

Valuation: remain EW, PT 185 NOK. Nordic Semi is well placed to capitalize on robust market growth and a favourable product cycle. We remain EW, as these positive trends seem priced in at current levels. Our price target of NOK 185 is based on 30x FY27 earnings, a slight premium to the 5 yr average 2 yr forward P/E multiple of 29x, reflecting cyclical momentum and the nRF54 platform introduction.

PDF Solutions - OW, PT \$62

Our thesis: Enabling the Semi Ecosystem. PDFS offers differentiated exposure to semis manufacturing, with structural complexity, a predominantly recurring revenue base and emerging AI monetisation supporting durable growth and margin expansion. We see backlog conversion, gross-margin recovery and tangible AI adoption as the key proof points for further upside.

Catalysts: We think 5 things will be monitored by investors from here. (i) AI monetisation: management's AI-agent narrative is strategically sound, but investors will look for measurable adoption, pricing or incremental revenue beyond positioning. (ii) Margin recovery: a 3Q gross-margin rebound would validate the mix explanation; slippage could raise questions around the economics of the hardware subscription model. (iii) Backlog conversion: continued H2 bookings and evidence that the \$271m backlog converts into sustained platform growth. (iv) DirectScan proof points: further mature-node production wins and progress from memory evaluation to commercial adoption. (v) PDFS Connect Conference 15-16 October 2026: investors will be looking for discussion on the latest developments to the Exensio platform.

Valuation: initiate at Overweight, PT \$62. Platform transition expands the opportunity as PDFS combines tools into a common semis data layer, increasing share, customer reach and durability of growth. Complexity is the structural growth engine, and we model a 21% sales CAGR (FY26–28e), with GM%/OP% reaching 77%/29% by FY28, broadly consistent with management's long-term framework. We apply a 35x P/E multiple to our FY27 EPS_e of \$1.77, in line with US Semi-Cap valuation and a premium to EDA multiples, reflecting its central position as an enabler of the semiconductor ecosystem.

See our initiation: [Enabling the Semi Ecosystem, Initiate at Overweight](#)

Soitec - OW, PT €200 (Top Pick)

Our thesis: Room for plenty more PICs in a POD. Soitec is leveraged to silicon photonics (SiPho) through its leadership in photonics SOI substrates, a critical input for optical networking modules. As SiPho adoption scales in AI datacenters, Soitec's opportunity increases. With near-term acceleration confirmed on both better demand visibility and execution, we see Soitec as a core holding for investors that want exposure to optical networking.

Catalysts: Our updated EPS estimates remain well ahead of the street: 28% and 49% on EPS for FY28e and FY29e. As such, we think further guidance raises similar to last week ([here](#)), would provide the confirmation the market is looking for on Soitec's photonics opportunity. Soitec is still valued as a mobile stock even though photonics provides c.55% of earnings next fiscal year, on our estimates, therefore any confirmation of the revenue opportunity and/or guidance uptick should support a rerating.

Valuation: remain OW, PT €200. We derive our €200 price target by applying a 25x multiple to our CY28e (fiscal '29 - March '29 year end) EPS estimate. We value Soitec on CY28e given increased visibility on the trajectory of growth for Photonics SOI, which we believe will be a multi-year growth story. Our multiple is in-line with sector peers.

STMicroelectronics - OW, PT €65

Our thesis: Optical and LEO upside amidst a cyclical recovery. We think the market

continues to overlook STMicro's structural growth vectors. On LEO, our checks suggest that satellite handset demand continues to be very strong and we expect this to continue to be supportive of growth into FY27. On optics, we note investor concerns on the sustainability of AI capex have halted networking sentiment. We see this as an opportunity to step in with many underappreciating the earnings potential from (i) optics and (ii) LEO (see more detail in our analog recap note [here](#)). On the cycle we think underlying dynamics remain strong, and we expect this to remain the case through into FY27 where we think MCU demand momentum and new product cycles will continue to support growth.

Catalysts: For rack architecture updates we now look to the AI Infra Summit in September and the OCP Global Summit in October. For networking specific commentary, we think the PIC Summit Europe in November may provide updates on CPO/NPO timelines and optical technology.

Valuation: retain Overweight, PT €65. We value STM at €65/share based on a sum-of-the-parts framework. We value the underlying core business at 15x P/E multiple on FY28e EPS €2.12 and the optical and LEO business on a 30x P/E multiple on FY28e EPS €1.33, to reflect the different growth trajectories of the cyclical and structural segments. Discounting using a 10% WACC, this implies a €65 PT.

Synopsys - OW, PT \$500

Our thesis: Positioned to benefit from Physical AI. We think opportunities in systems simulation present a multi-year growth driver, especially with the emergence of physical AI. We think Design IP recovery looks increasingly convincing, especially given the new Factory 2 IP model.

Catalysts: Synopsys holds its investor day on 30th September. At the last print management circled on two points of focus: (i) multi-physics fusion technology and (ii) the factory 2 IP model. While we appreciate the former plays into simulation with the latter supporting design IP growth, we think the market now has high expectations for this investor day with the stock having traded up 11% from the last print ([link](#)). Our sense is that the market is looking for evidence of better than previously guided to Ansys revenue synergies from the multi-physics technology, as well as a better Design IP guide from the factory 2 IP model.

Valuation: we move to OW, PT \$500. Valuation looks undemanding relative to peers as while we agree Cadence should trade at a premium to Synopsys, we think this gap has now become too wide. Synopsys currently trades on a 28x P/E multiple on this year's earnings vs Cadence on a 37x P/E multiple. We continue to value Synopsys on a 33x P/E multiple at the midpoint of the historical trading range of the stock, 30-35x. Applying this multiple to our FY26 EPS of \$15.12 sets our PT at \$500.

VAT Group - EW, PT CHF 625

Our thesis: positioned to benefit from strong WFE growth, but we see better value elsewhere and remain Equal-weight. VAT is a well run business with a strong market position and good growth prospects, but so is the rest of our semi-cap coverage. VAT's 2018-28e revenue CAGR is broadly in line with the top six WFE companies and its margin

profile is, if anything, slightly below them currently.

Valuation: remain EW, PT CHF 625. We stay EW on VAT as this group trades on around 31x FY27 earnings against VAT at 43x our FY27 EPS, on similar growth drivers. Our price target is derived by applying a P/E multiple of 40x to estimated FY27 earnings. This aligns with the 5-year average NTM P/E of c.41x but sits well above the semi-cap group at 31x FY27. Although a premium is warranted due to CHF's safe haven status and a strong local bid, we are EW as we see better value elsewhere.

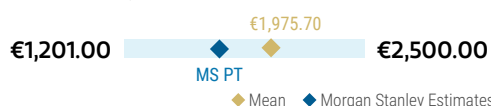
Risk Reward – ASML Holding NV (ASML.AS) Top Pick

DRAM and Logic Demand Accompanied by Margin Expansion

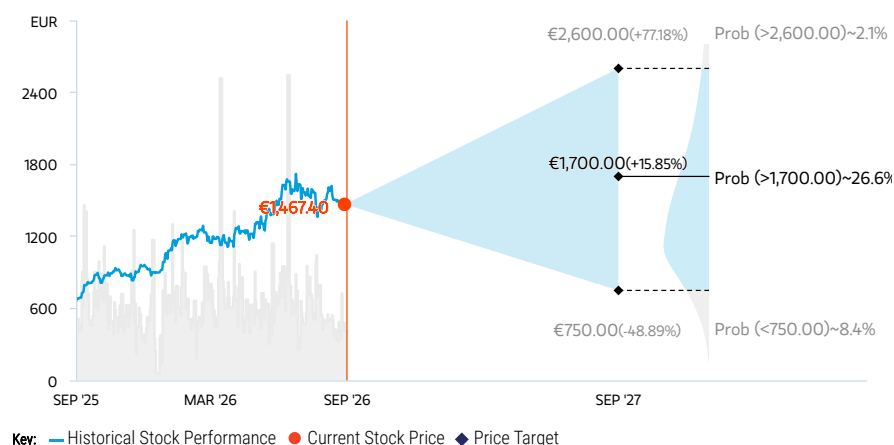
PRICE TARGET €1,700.00

We value ASML at a P/E multiple of 30x, at the lower end of our usual cycle peak valuation range of 30-35x. This is in line with historical peaks, but we note upside potential from significant foundry capex as well as continued rapidly rising memory prices. Applying a 30x P/E multiple to our €56.42 FY28 EPS_e gives us a PT €1,700.

Consensus Price Target Distribution



RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)

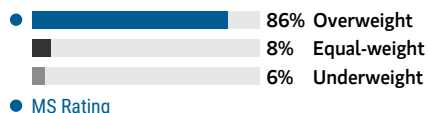


Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 4 Sep 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology [here](#)

OVERWEIGHT THESIS

- ASML is a dominant supplier of critical toolsets for the semi manufacturing process.
- We expect WFE growth, although we see lithography intensity (share of WFE) flattening over 2024-26, which may cap earnings in the meantime.
- We anticipate consensus earnings revisions as debate shifts to 2026-27 on improving DRAM and logic demand for EUV.
- We anticipate better than expected customer demand for holistic lithography solutions.
- Metrology margins look robust as use of eBeam drives growth and near universal adoption.
- We expect supply chain management to deliver growing GM% across the mid to longer term.

Consensus Rating Distribution



Risk Reward Themes

Technology Diffusion: *Positive*

View descriptions of Risk Rewards Themes [here](#)

BULL CASE €2,600.00

DRAM and 2nm/A16 logic rapid growth
c.40x 2028e bull case EPS c.€65: Rapid increase in orders with successful technology node transitions requiring broad-based capacity expansions. This is supported by large data centre buildout and big logic capacity build. Limited impact from export controls, seamless adoption of High NA and low capital intensity sees GM% trending towards 60% by FY28. This sets our FY28 bull case EPS at €65. We apply a c.40x P/E multiple to reach €2,600.

BASE CASE €1,700.00

Solid DRAM demand accompanied by margin expansion
c.30x on 2028e EPS €56.42: We model recovery from the leading foundry customer, solid DRAM (for HBM) and China demand as better than feared. This is in line with historical cycle peaks although this may go a little higher given the significant foundry capex as well as continued soaring memory prices. Risks to estimates include (i) higher than expected immersion demand pull-in (China), (ii) slow EUV order book expansion and/or (iii) push-out of initial High NA sales.

BEAR CASE €750.00

Semi industry downturn emerges
c.20x 2028e bear case EPS c.€37.5: Equipment market recovery in advanced nodes is weak as equipment demand stalls, leading ASML to witness a year of weak industry spend. Poor sales expansion with some development pull-ins lends itself to a lower EPS.

Risk Reward – ASML Holding NV (ASML.AS)

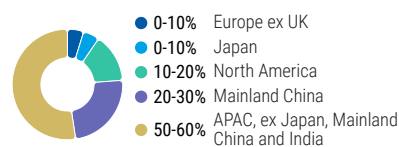
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Sales growth - yy (€, mn)	0	0	0	0
Gross Margin (GAAP) (€, mn)	1	1	1	1
Operating Margin (GAAP) (€, mn)	0	0	0	0

INVESTMENT DRIVERS

- 3/2nm transitions on logic/foundry & 1a/b transition on memory drive demand for tools
- Expand EUV margin to DUV levels (>50%)
- Higher throughput drives: (1) stronger case for use in DRAM; (2) higher service revenue linked to output, not maintenance

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

2/5
MOST

3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Meaningful ramp-up in logic/foundry spend with nrs 2 and 3 returning to investment mode
- Higher HBM/DRAM spend than expected
- Servicing margins expanding
- Tech sovereignty programmes crystallising

RISKS TO DOWNSIDE

- Materially weaker end-market demand (foundry / DRAM especially)
- Strong inflation cycle and Chinese weakness driving slowdown and more order push-outs
- Absent High NA adoption (DRAM 2028+) inhibiting multiple

OWNERSHIP POSITIONING

Inst. Owners, % Active	74.9%				
HF Sector Long/Short Ratio	2.6x				
HF Sector Net Exposure	14.3%				

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e

Sales / Revenue (€, mn)	47,576	50,887	55,579	58,983
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EBIT (€, mn)	18,577	21,392	24,085	26,782
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EPS (€)	37.26	49.00	53.46	59.10
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◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Infineon Technologies AG (IFXGn.DE)

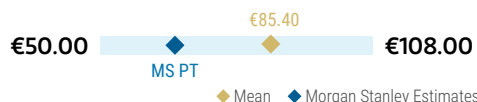
Structural Breakout Story Playing Out But Priced In

PRICE TARGET €65.00

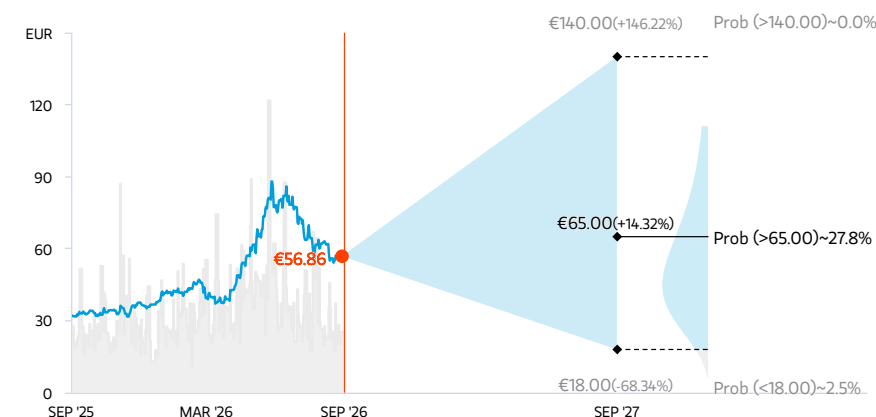
We apply a 30x P/E multiple to the server rack contribution as we think the structural driver deserves a premium to the rest of the group. We continue to view the rest of the group as a cyclical story valuing it at 20x reflecting our view that Infineon is moving through a cycle recovery. Using a SOTP and assuming a 10% WACC, we value IFX at €65.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)



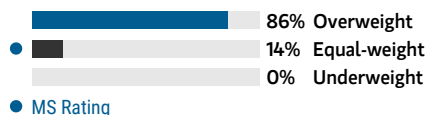
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 4 Sep 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology [here](#)

EQUAL-WEIGHT THESIS

- We view Infineon as best placed amongst European analog names for a cyclical recovery in automotive/industrial, demonstrating a superior ability to weather the headwinds facing these markets in the meantime.
- We expect new wins in autos to validate IFX's strategy (design focus, capacity build, proprietary IP). A strong IP library augurs well for GaN longer term too including use in EVs, data centres and energy storage.
- We think Infineon will be the clear data center beneficiary with the full suite of power semi solutions for each stage of the rack.
- However near-term share price movements will depend on clarification of the 800v deployment timeline.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Electric Vehicles:	Positive
Renewable Energy:	Positive
Secular Growth:	Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

€140.00

Solid auto growth and v-shaped economic recovery

65x P/E for the structural breakout story and 25x for the cyclical story

Scenario driven by faster than expected recovery in automotive production, especially electric vehicles, as well as faster migration to SiC based designs and higher level ADAS adoption (level 2+). Meaningful recovery in industrial through FY25. Segment margin pushes towards c.30% and new capacity adds place the company well for further expansion over the longer term. Non-GAAP EPS of €4.50 FY28e based on €25bn sales.

BASE CASE

€65.00

Cyclical Recovery and Server Rack Breakout 30x Server rack breakout story and 20x for the cyclical story

Due to FX and tariff headwinds, FY25 sales were down c.2% for the year. We forecast c.12% growth in FY26e, 15% growth in FY27e and 13% growth in FY28e. We model non-GAAP GM% (FY25: 41.4%) to grow to 43.3% in FY26e, 46.9% in FY27e and 49.3% in FY28e as greater autos & industrial demand along with increased utilisation levels take effect. This results in €1.75 FY26e EPS, €2.86 FY27e EPS and €3.61 FY28e EPS.

BEAR CASE

€18.00

Macro weakness and lower design wins

No structural breakout assumed so use P/E c.12x EPS of €1.50

Macro weakness is worse than expected with end demand remaining sluggish in both automotive and industrial for a prolonged period. Segment margins drop to high-teens% on higher under-utilisation charges, which drag as inventory builds up within the semi industry channels. Design win momentum with SiC stalling as does ADAS adoption. FY26e EPS fall to c.€1.5.

Risk Reward – Infineon Technologies AG (IFXGn.DE)

KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Revenue Growth (%)	(2)	12	15	13
Segment Margin (%)	17.5	19.5	25.1	27.4
Adjusted EPS - Diluted (€)	1.39	1.75	2.86	3.61

CATALYST CALENDAR

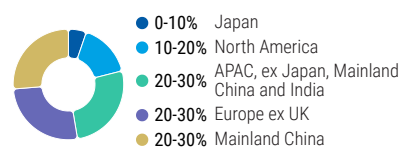
Date	Event
10 Nov 2026	Q4 2026 Infineon Technologies AG Earnings Release
02 Feb 2027 - 08 Feb 2027	Q1 2027 Infineon Technologies AG Earnings Release
19 Feb 2027 - 23 Feb 2027	Infineon Technologies AG Annual Shareholders Meeting

Source: Refinitiv, Morgan Stanley

INVESTMENT DRIVERS

- Adoption curve of EV & ADAS technology in passenger vehicles
- Successful execution of SiC strategy in power semis; IP bolstering add-ons in GaN
- Capacity buildout sensibly balances near-term macro downturn and long-term potential

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

2/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Faster than expected recovery of industrial output & consumer sales
- Longer autos semi cycle than normal & better pricing power with OEMs
- Greater deployment of power semis to the data centre than expected

RISKS TO DOWNSIDE

- Sustained weakness in global automotive sales and/or PMIs
- Capex ramp leads to over-capacity and impairs future returns
- Backlog slowdown
- Chinese designs limiting longer term share expansion

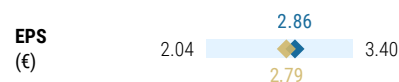
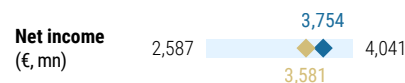
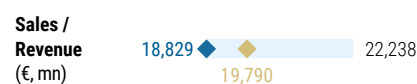
OWNERSHIP POSITIONING

Inst. Owners, % Active	67.5%				
HF Sector Long/Short Ratio	2.6x				
HF Sector Net Exposure	14.3%				

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Dec 2027e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – Synopsys Inc. (SNPS.O)

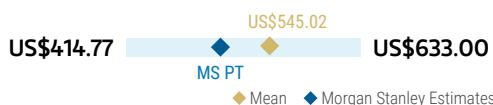
Positioned to benefit from Physical AI

PRICE TARGET US\$500.00

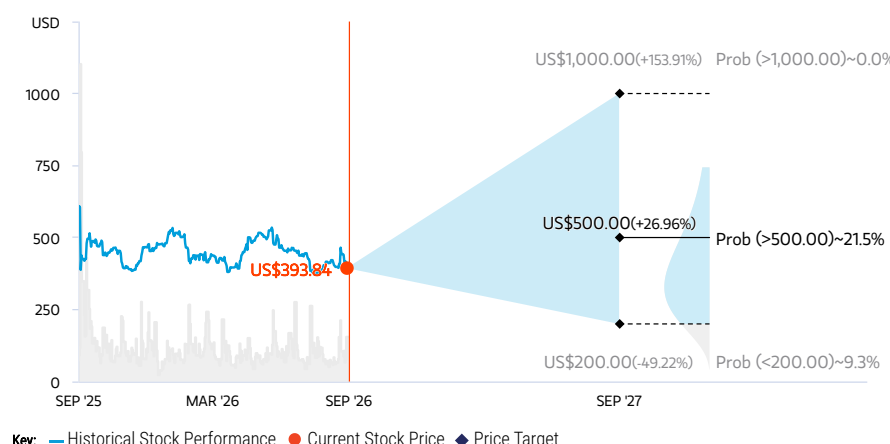
The sector is trading on a 30-35x P/E multiple range. We expect Synopsys to trade in line and apply a c.33x P/E multiple to estimated FY26 EPS (\$15.12).

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)



Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 4 Sep 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology [here](#)

BULL CASE US\$1,000.00

Accelerating HPC chip development & security test

Synopsys sees better gains than modelled in AI as design range and complexity expand through the rest of this decade at a faster pace than expected. Edge AI and new autos compute (ADAS, SDV driven) add further momentum. Software testing finds a large, niche market (security?) and moves to corporate average margins within 3 years. We estimate Synopsys could reach c.\$25 EPS on such accelerating development trends, which on a cycle high of c.40x could see a c.\$1,000 share price.

BASE CASE US\$500.00

Multi-year growth drivers

We think opportunities in systems simulation present a multi-year growth driver, especially with the emergence of physical AI. We think Design IP recovery looks increasingly convincing, especially given the new Factory 2 IP model. We model EPS of \$15.12 by FY26 and, applying a 33x P/E multiple, we reach a \$500 base case valuation.

BEAR CASE US\$200.00

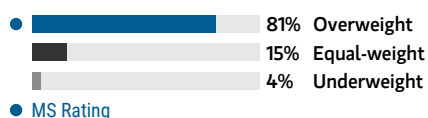
Slowing AI development as macro trumps secular

Assumes that demand for Synopsys' tools and IP wanes as the down-cycle in Semis becomes deep enough even to directly affect strategic spend (R&D) in design flow. Chip development is postponed at key customers as macro weakness and weak demand visibility continue, dampening sales growth to low single digits, with support mainly from high recurring revenues. In this scenario, we see FY26 EPS at c.\$8, which on a cycle trough of c.25x could see the share price pared back to c.\$200.

OVERWEIGHT THESIS

We think opportunities in systems simulation present a multi-year growth driver. We assume a 1-year forward P/E multiple range of 30-35x in setting our price target of \$500. We are Overweight as we think (i) Design IP recovery is becoming more evident, (ii) Ansys synergies are materialising and (iii) the simulation portfolio is positioned to benefit from the emergence of physical AI.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

New Data Era: *Positive*
 Secular Growth: *Positive*
 Technology Diffusion: *Positive*

View descriptions of Risk Rewards Themes [here](#)

Risk Reward – Synopsys Inc. (SNPS.O)

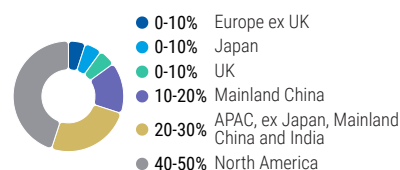
KEY EARNINGS INPUTS

Drivers	Oct 2025	Oct 2026e	Oct 2027e	Oct 2028e
Sales growth - yy (%)	15.1	37.9	10.5	10.6
Gross margin (%)	77.0	73.0	73.7	73.8
EBIT margin (reported) (%)	13.0	9.8	15.2	17.9

INVESTMENT DRIVERS

- Attractive portfolio positioned for growth (AI chip development)
- Sustained earnings leverage on double-digit EDA sales CAGR
- Structural opportunities including Physical AI
- Automotive and industrial recovery
- Joint Synopsys-Ansys products

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

4/5 BEST	24 Month Horizon	2/5 MOST	3 Month Horizon
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Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Bigger ramp-up in AI ASICs and GPUs than modelled
- Quicker recovery in Automotive & industrial
- Design wins at expense of peers with Ansys products
- Strong recovery in IP business (esp. in China)

RISKS TO DOWNSIDE

- Scale and duration of downturn seeing deeper R&D cuts
- Further tightening of export controls
- Loss of share to new entrant EDA players given localisation trends
- Failure to hit cost synergies from Ansys merger

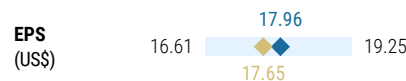
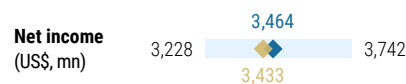
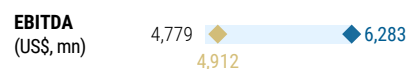
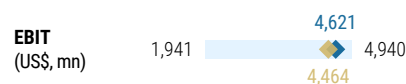
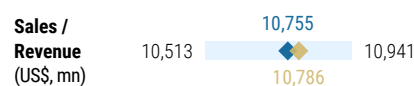
OWNERSHIP POSITIONING

Inst. Owners, % Active	51.7%				
HF Sector Long/Short Ratio	2.1x				
HF Sector Net Exposure	33.6%				

Refinitiv; MSPB Content. Includes certain hedge fund exposures held with MSPB. Information may be inconsistent with or may not reflect broader market trends. Long/Short Ratio = Long Exposure / Short exposure. Sector % of Total Net Exposure = (For a particular sector: Long Exposure - Short Exposure) / (Across all sectors: Long Exposure - Short Exposure).

MS ESTIMATES VS. CONSENSUS

FY Oct 2027e



◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

Risk Reward – BE Semiconductor Industries NV (BESI.AS)

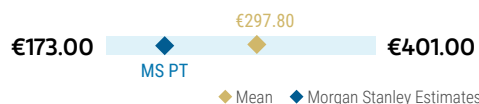
Mainstream to stay stronger for longer, but HBM likely to come later and lower.

PRICE TARGET €220.00

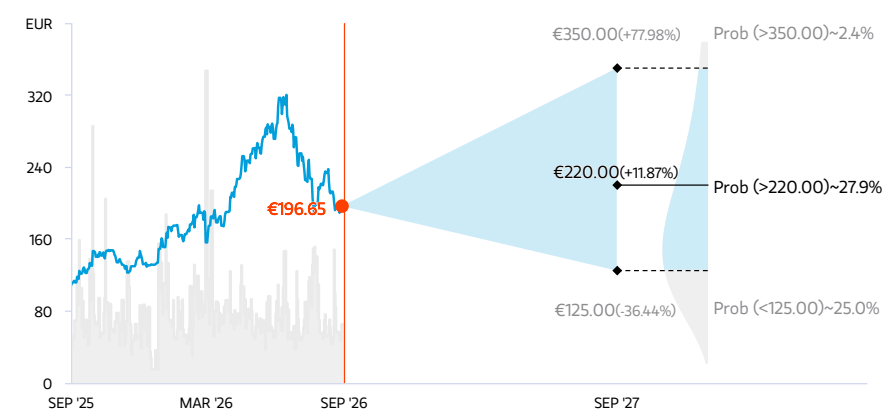
We apply a 31x multiple to FY27 earnings power which is at the lower end of the company's 5 year average P/E multiple range of c.30-35x and reflects the emergence of risks to HB adoption.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART AND OPTIONS IMPLIED PROBABILITIES (12M)



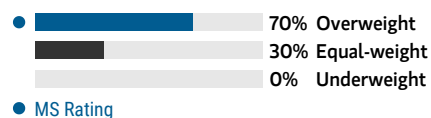
Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research, Morgan Stanley Institutional Equities Division. The probabilities of our Bull, Base, and Bear case scenarios playing out were estimated with implied volatility data from the options market as of 4 Sep 2026. All figures are approximate risk-neutral probabilities of the stock reaching beyond the scenario price in either three-months' or one-years' time. View explanation of Options Probabilities methodology [here](#)

OVERWEIGHT THESIS

- The set-up for Besi's core business continues to improve, evidenced by a need for chiplet architectures (2.5D) for AI accelerators
- Accelerating hybrid bonding adoption in AI (both compute and networking) has started to result in meaningful orders
- Valuation in the high 20s FY27e P/E is not reflecting optionality of broadening hybrid bonding adoption outside of HBM

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Disruption:	Positive
New Data Era:	Positive
Secular Growth:	Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

€350.00

Broad strength on 40x P/E

Our bull case reflects more optimism across the board. For the mainstream business we expect growth to be higher for longer. For TCB we expect Besi to gain a meaningful foothold while for HB we expect further upside from accelerated adoption, including in HBM. As a result our bull case EPS estimate is c.€9, 25% ahead of our base case, to which we apply a 40x multiple to derive a €350 fair value estimate.

BASE CASE

€220.00

Clear evidence of an AI infused (super) cycle

Our base (31x estimated FY27 EPS) reflects strengthening momentum in the core business and acceleration in the adoption of hybrid bonding. Given Besi's industry-leading margin profile, we expect that the company should be able to generate strong earnings power. The multiple we apply is at the lower end of the stock's 5yr average 30-35x range.

BEAR CASE

€125.00

Slower growth across the board

Our bear case reflects slower growth in the mainstream business and relatively lacklustre HB/TCB growth compared to our base case estimate resulting in c.30% lower estimates. We apply a 25x multiple to our FY27 EPS estimate to calculate a fair value of €125 in this scenario.

Risk Reward – BE Semiconductor Industries NV (BESI.AS)

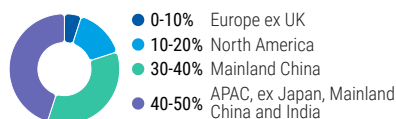
KEY EARNINGS INPUTS

Drivers	Dec 2025	Dec 2026e	Dec 2027e	Dec 2028e
Revenue growth (%)	(3)	67	43	9
EBIT margin (%)	29	42	47	49
FCF	133	321	508	617

INVESTMENT DRIVERS

- Observed strength by Besi and peers in packaging applications
- Announcements by companies adopting hybrid bonding
- Adoption of TCB by customers

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies [here](#)

MS ALPHA MODELS

4/5
MOST
3 Month
Horizon

Source: Refinitiv, FactSet, Morgan Stanley Research; 1 is the highest favored Quintile and 5 is the least favored Quintile

RISKS TO PT/RATING

RISKS TO UPSIDE

- Faster adoption of hybrid bonding in HBM memory applications
- Faster and stronger acceleration of Besi's core business
- Higher multiple due to (perceived) structural elements

RISKS TO DOWNSIDE

- Hybrid bonding adoption more limited than expected
- Strength in core business fades
- FX headwinds pressure margins

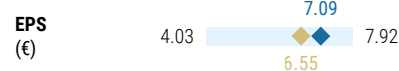
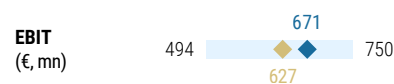
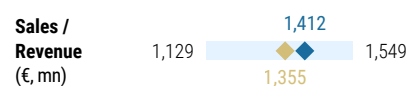
OWNERSHIP POSITIONING

Inst. Owners, % Active	80.5%	
HF Sector Long/Short Ratio	2.6x	
HF Sector Net Exposure	14.3%	

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MS ESTIMATES VS. CONSENSUS

FY Dec 2027e



◆ Mean ◆ Morgan Stanley Estimates
Source: Refinitiv, Morgan Stanley Research

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - [Options_Probabilities_Exhibit_Link.pdf](#)
2. View descriptions of Risk Rewards Themes - [RR_Themes_Exhibit_Link.pdf](#)
3. View explanation of regional hierarchies - [GEG_Exhibit_Link.pdf](#)
4. View explanation of Theme/Exposure methodology - [ESG_Sustainable_Solutions_External_Link.pdf](#)
5. View explanation of HERS methodology - [ESG_HERS_External_Link.pdf](#)

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(as of August 31, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1555	42%	450	48%	29%	767	44%
Equal-weight/Hold	1577	43%	395	42%	25%	770	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	559	15%	93	10%	17%	216	12%
Total	3,694		939			1754	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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INDUSTRY COVERAGE: Technology - European Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (09/07/2026)
Lee Simpson		
ASML Holding NV (ASML.AS)	O (09/22/2025)	€1,500.40
Infiniteon Technologies AG (IFXGn.DE)	E (09/08/2026)	€60.79
STMicroelectronics NV (STMPA.PA)	O (03/26/2026)	€46.64
Nigel van Putten		
Aixtron SE (AIXGn.DE)	E (05/25/2023)	€38.98
ASM International NV (ASMI.AS)	O (06/19/2024)	€862.20
BE Semiconductor Industries NV (BESI.AS)	O (11/07/2022)	€202.50
Melexis N.V. (MLXS.BR)	E (02/05/2026)	€67.85
Nordic Semiconductor ASA (NOD.OL)	E (02/10/2025)	NKr 172.00
Soitec SA (SOIT.PA)	O (03/26/2026)	€143.35
VAT Group AG (VACN.S)	E (03/21/2025)	SFr 629.60

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* Historical prices are not split adjusted.